St. Martins Marsh Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

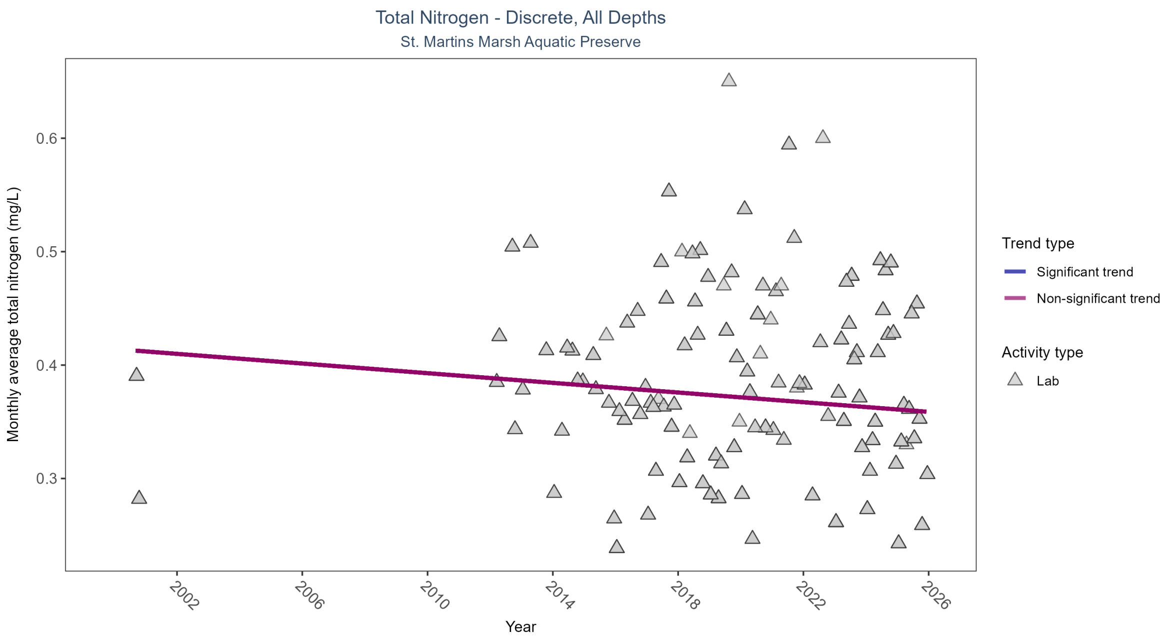

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	673	15	2000 - 2025	0.365	-0.08575	0.41416	-0.00213	0.1966

Total nitrogen showed no detectable trend between 2000 and 2025.

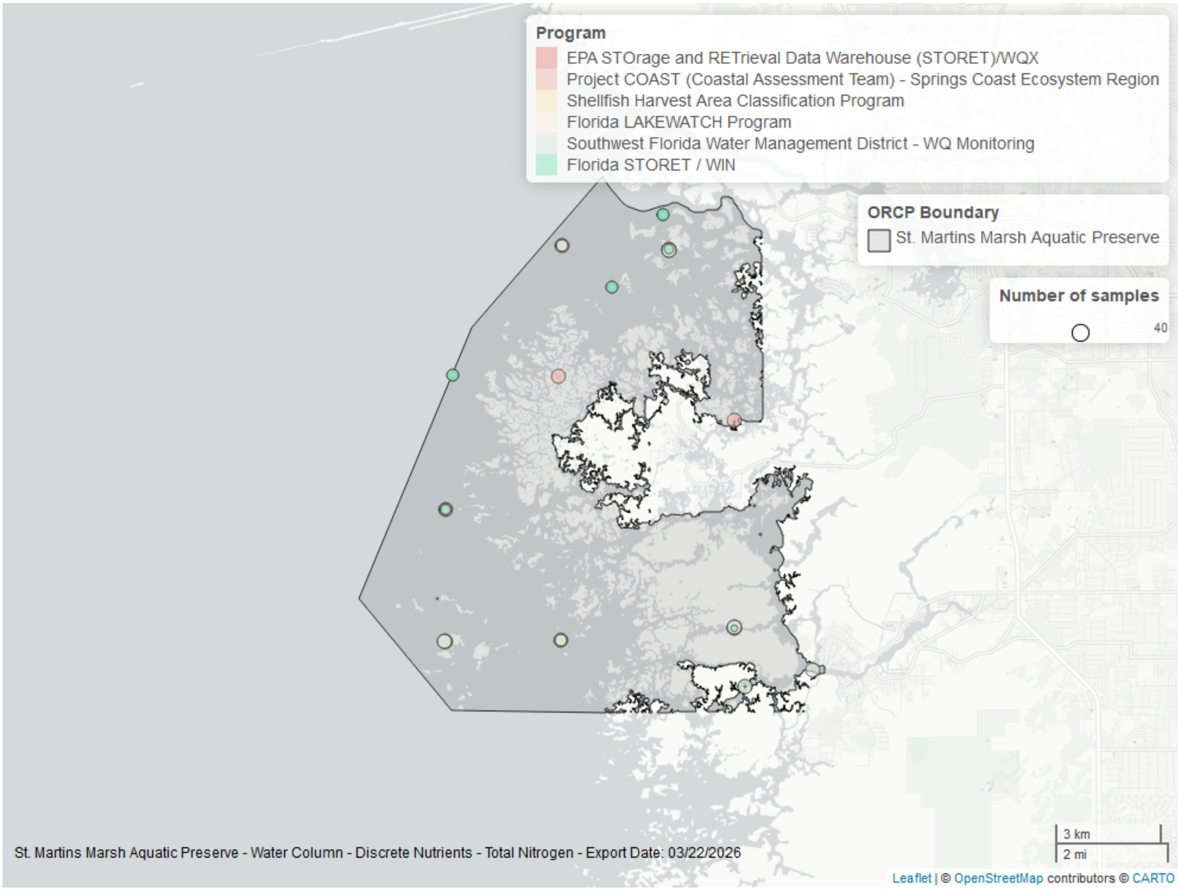

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

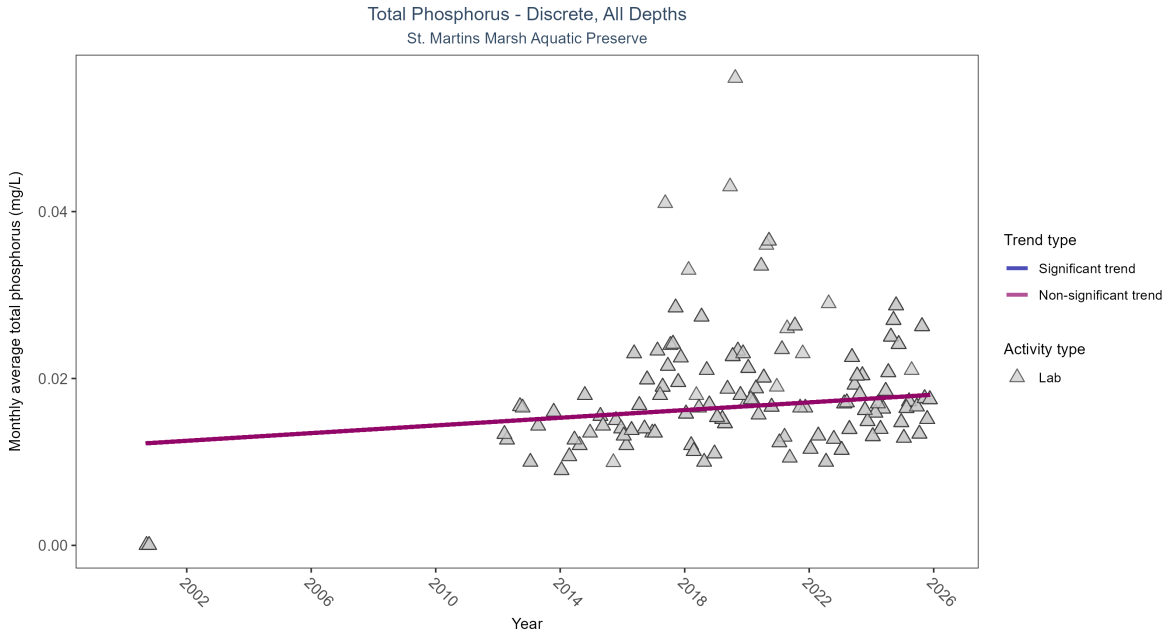

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	663	15	2000 - 2025	0.016	0.0941	0.01207	0.00023	0.1113

Total phosphorus showed no detectable trend between 2000 and 2025.

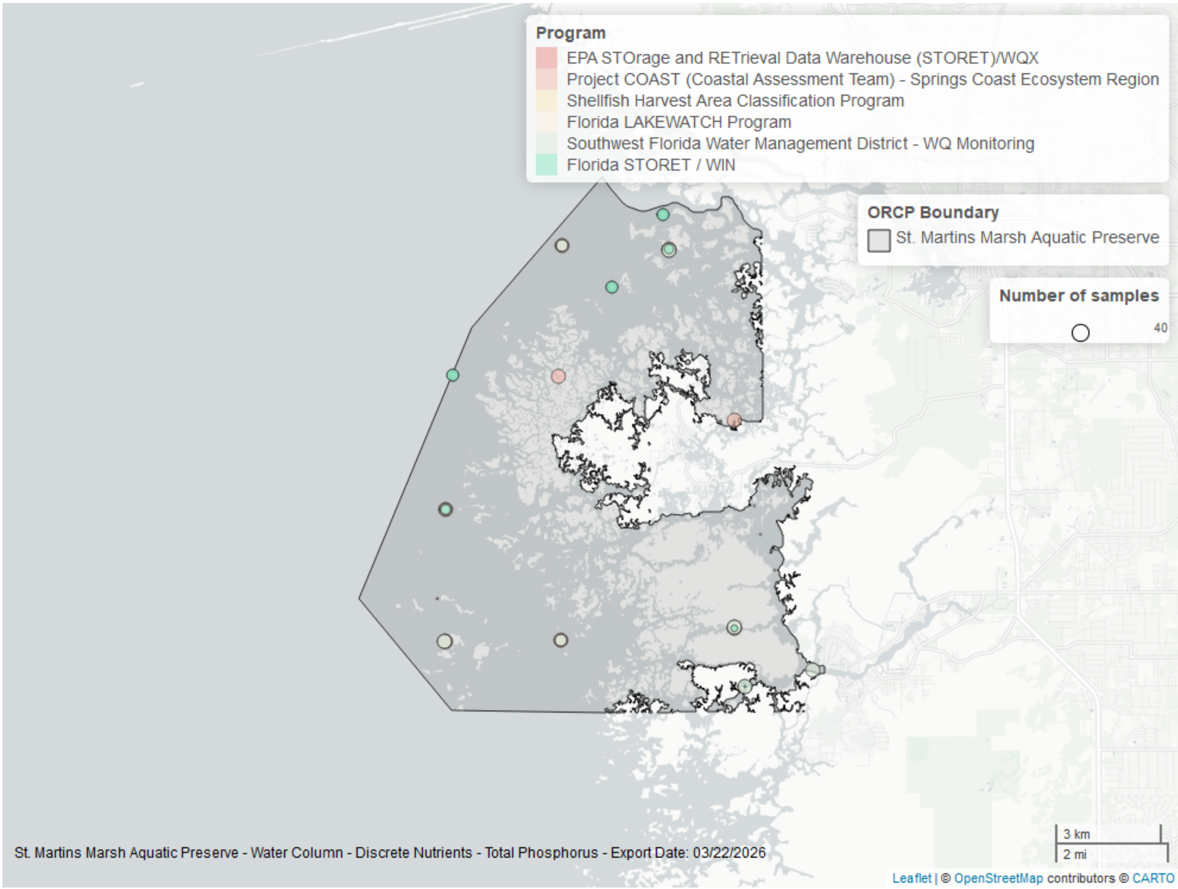

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

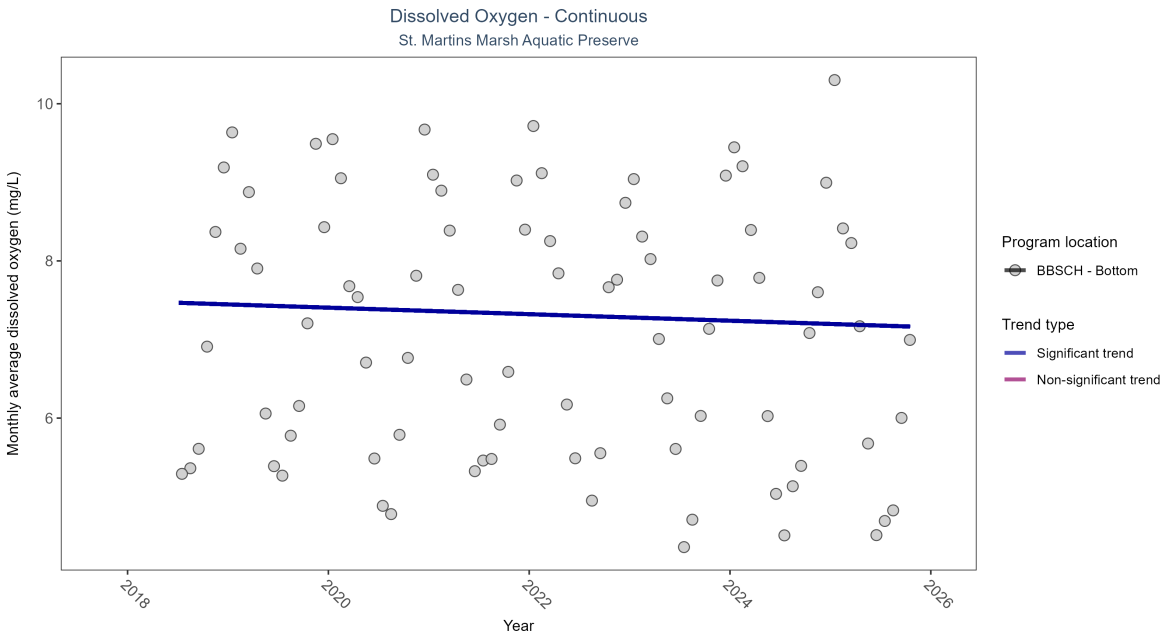

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly decreasing trend	227936	8	2018 - 2025	7.2	-0.21	7.49	-0.04	0.0217

At one program location, monthly average dissolved oxygen decreased by 0.04 mg/L per year.

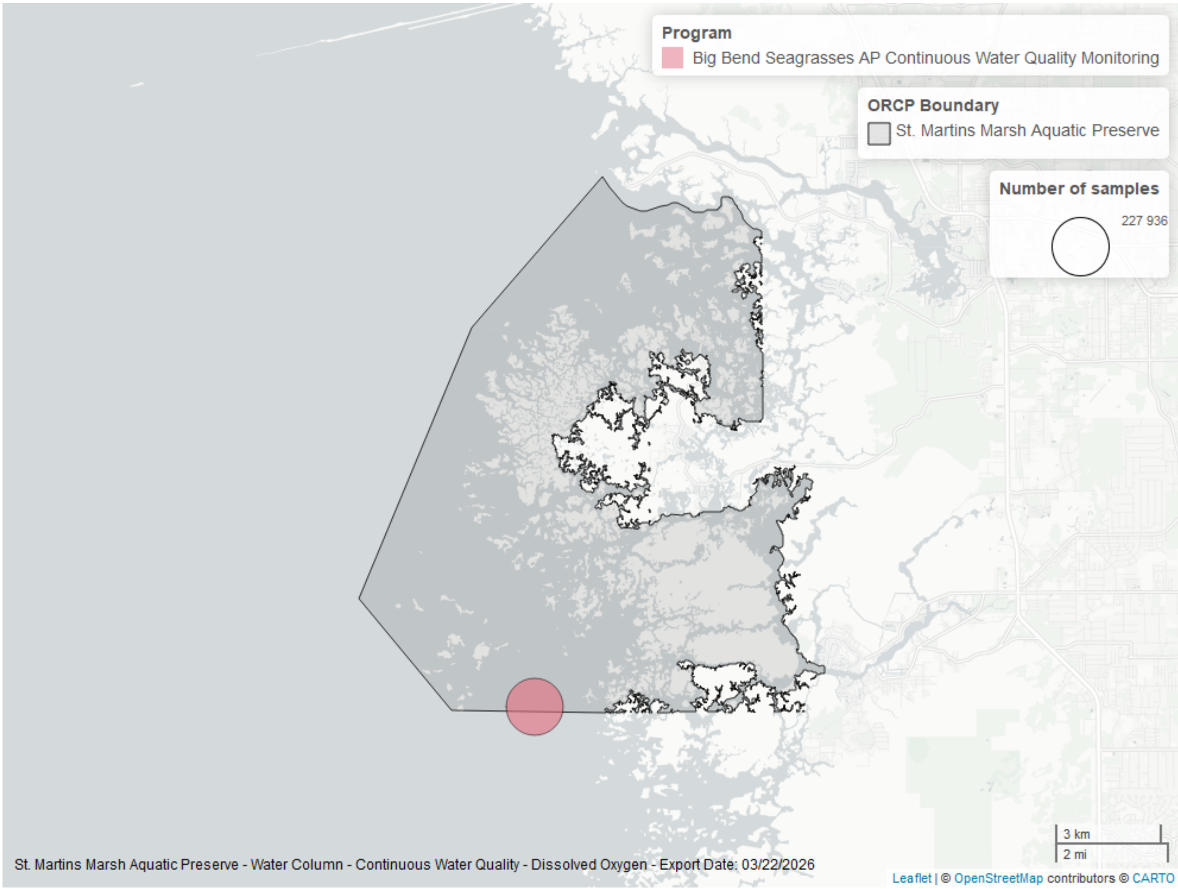

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

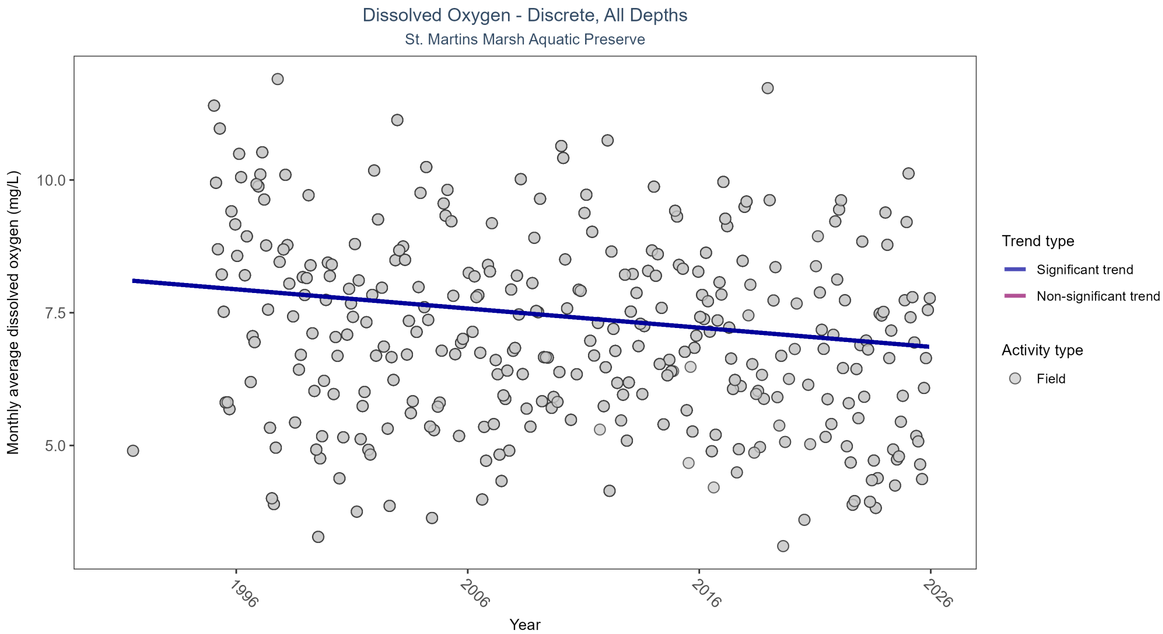

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	7885	32	1991 - 2025	7.3	-0.19439	8.12148	-0.03613	0

Monthly average dissolved oxygen decreased by 0.04 mg/L per year.

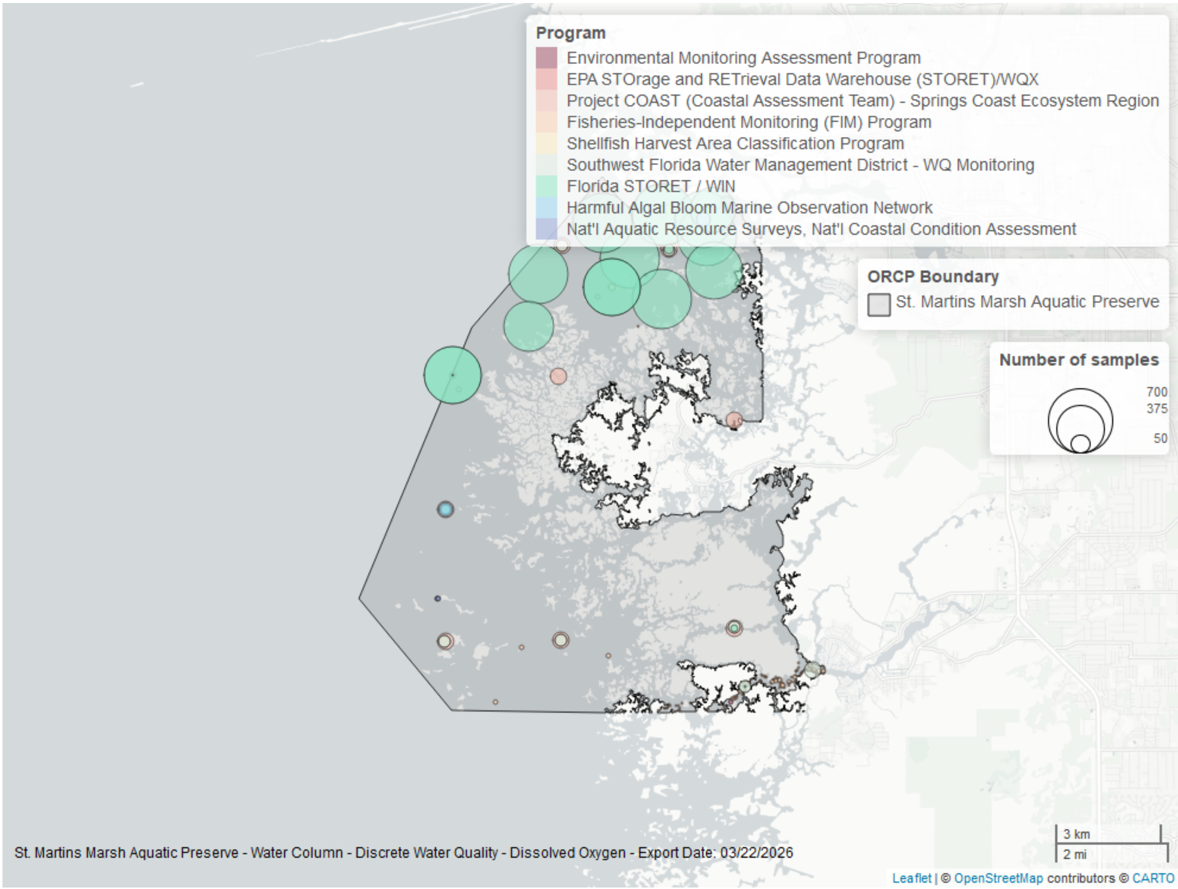

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

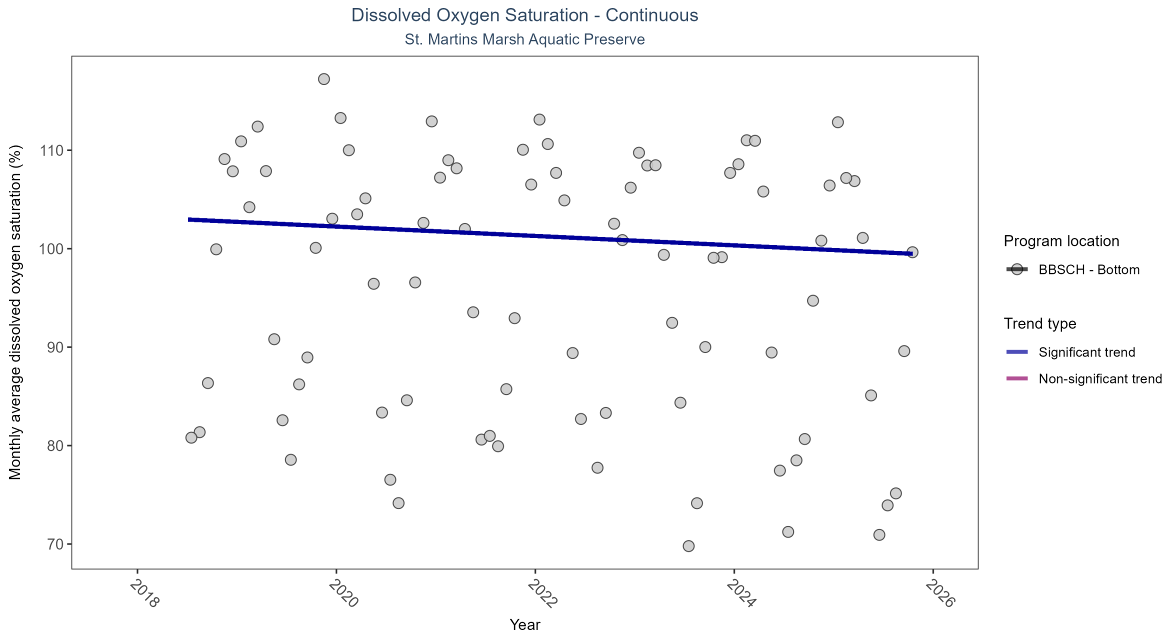

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly decreasing trend	227936	8	2018 - 2025	97.7	-0.26	103.2	-0.48	0.0041

At one program location, monthly average dissolved oxygen saturation decreased by 0.48% per year.

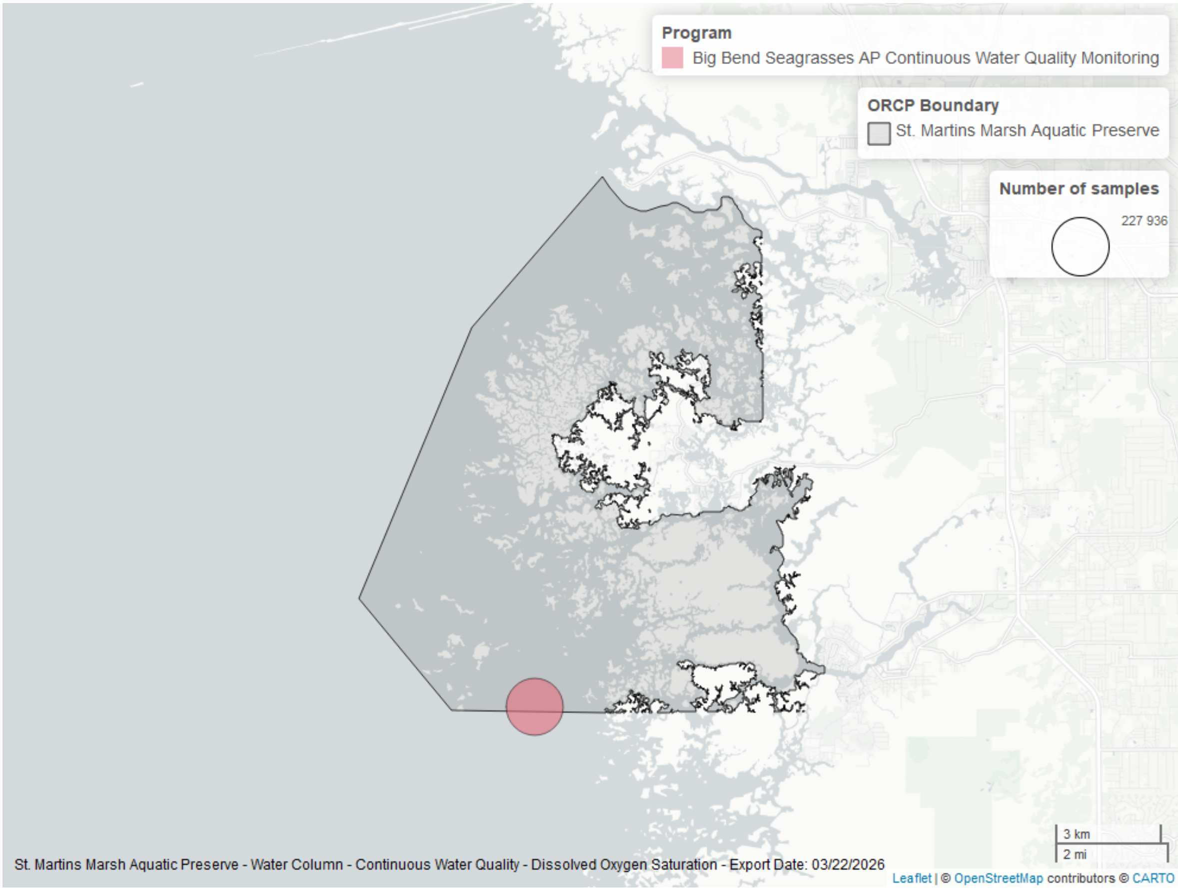

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

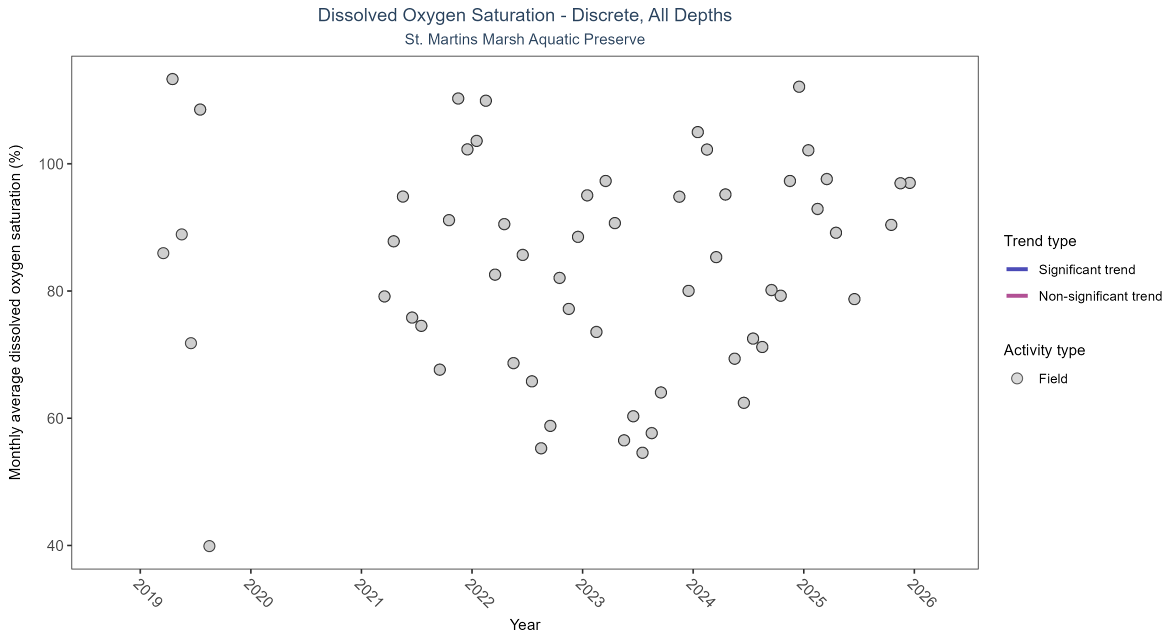

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data to calculate trend	461	6	2019 - 2025	86.6	-	-	-	-

There was insufficient data to fit a model for dissolved oxygen saturation.

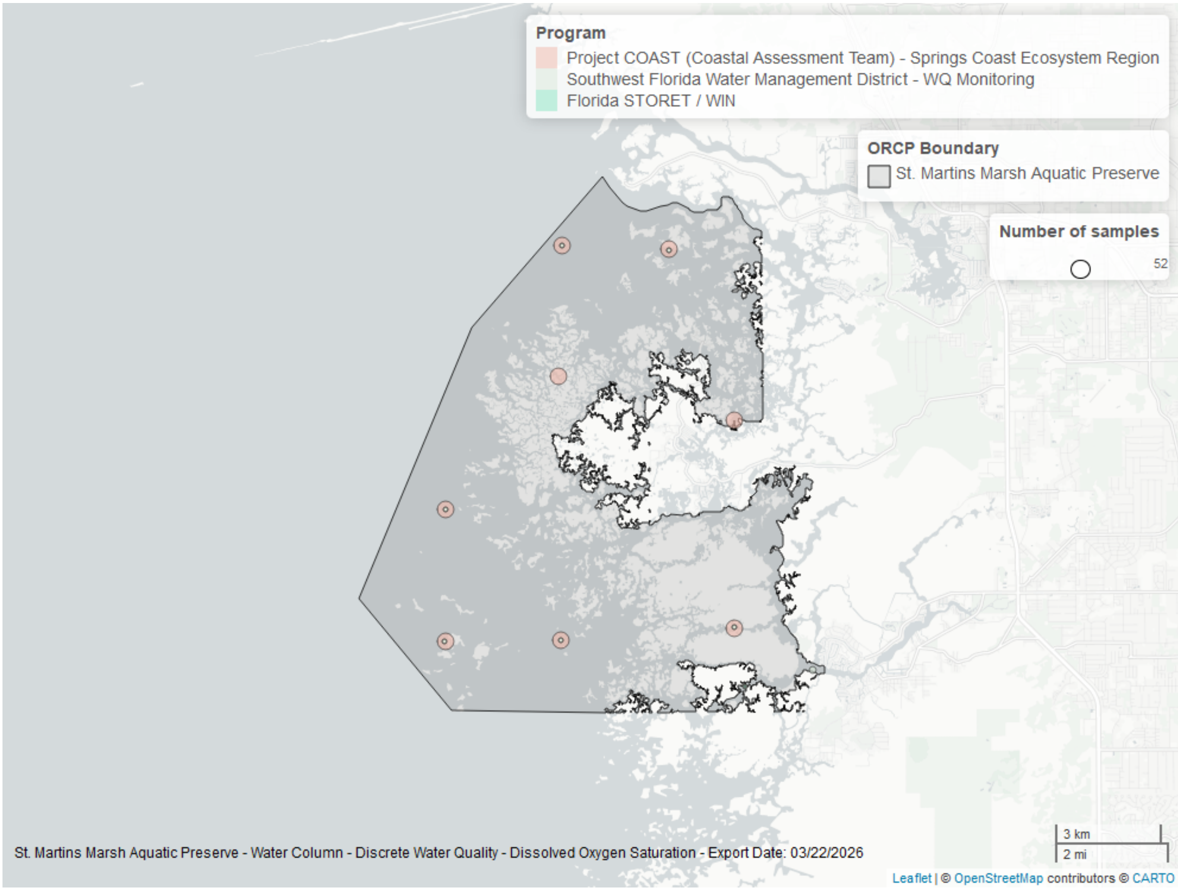

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

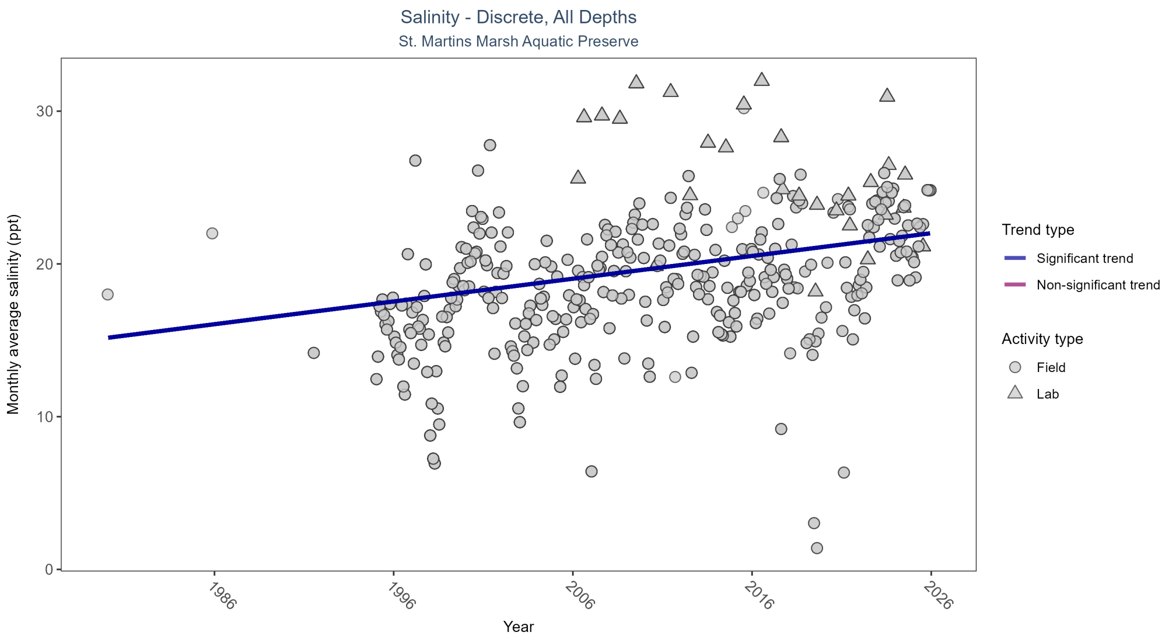

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly increasing trend	9666	34	1980 - 2025	18.4	0.25999	15.158	0.14896	0

Monthly average salinity increased by 0.15 ppt per year.

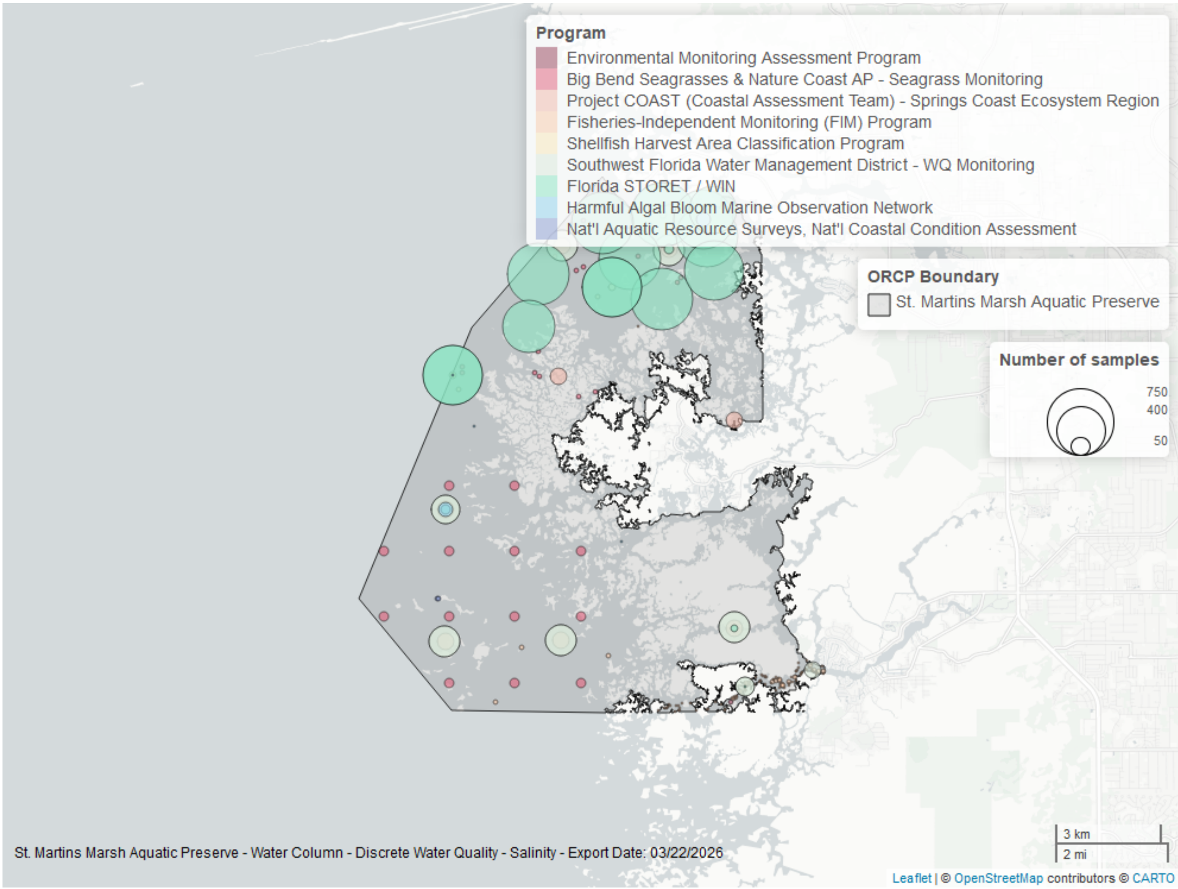

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

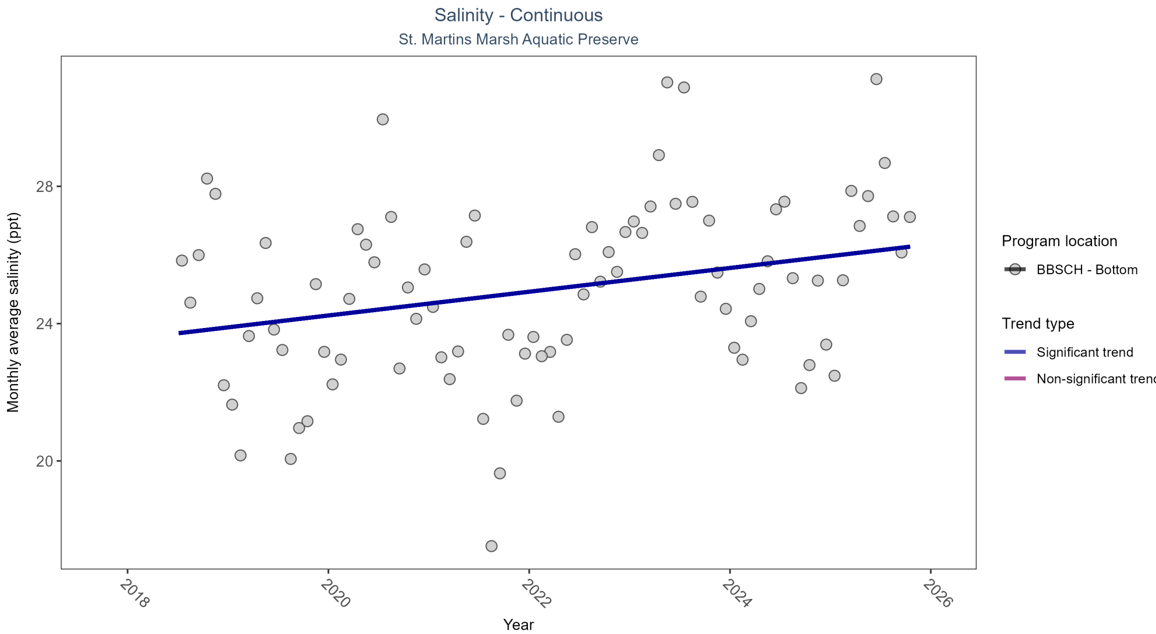

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly increasing trend	235777	8	2018 - 2025	25.2	0.29	23.55	0.35	0.0015

At one program location, monthly average salinity increased by 0.35 ppt per year.

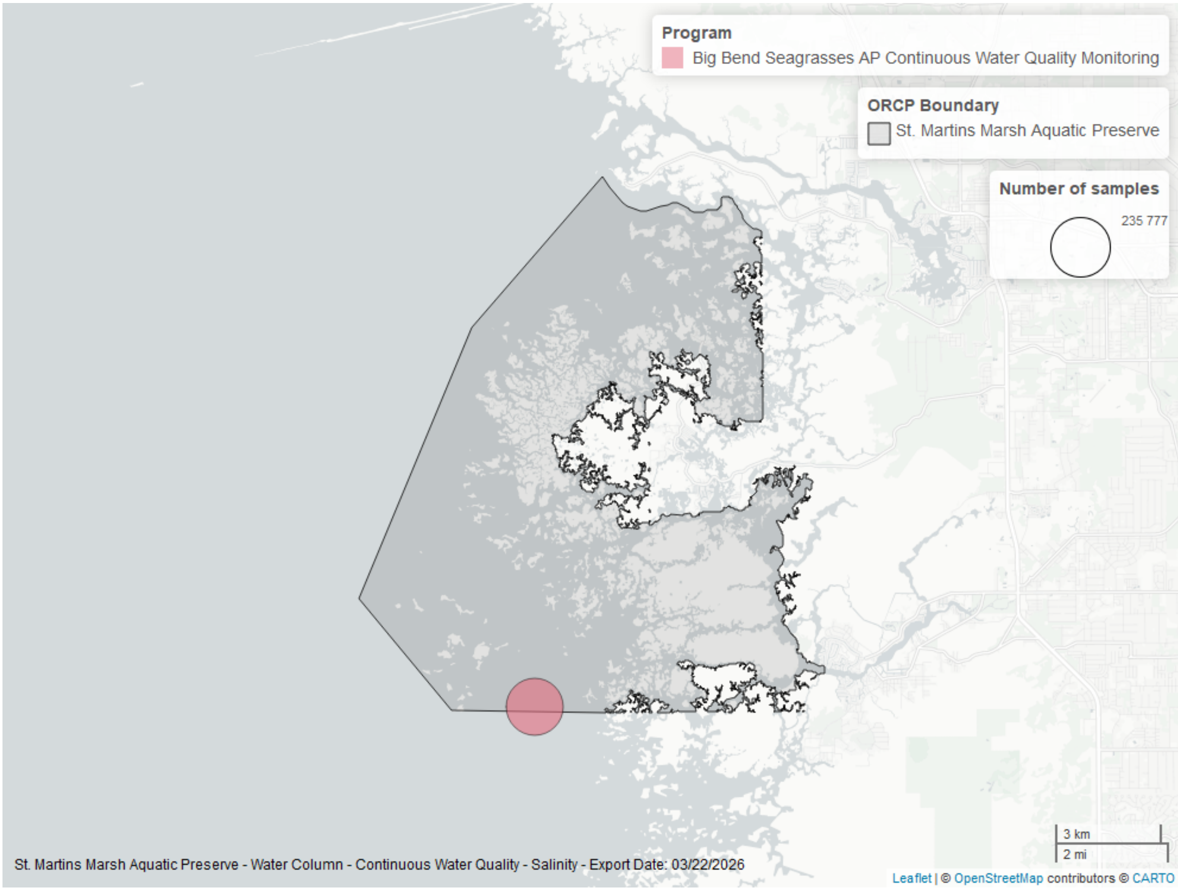

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

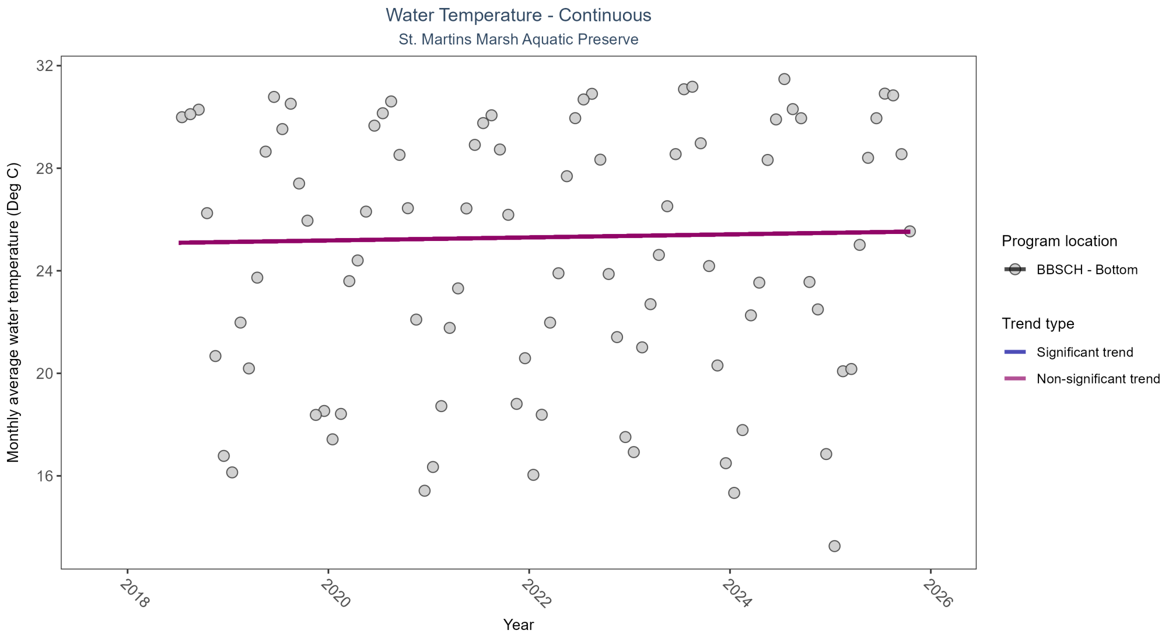

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	No significant trend	237837	8	2018 - 2025	25.3	0.06	25.06	0.06	0.4934

No detectable change in monthly average water temperature was observed at one location.

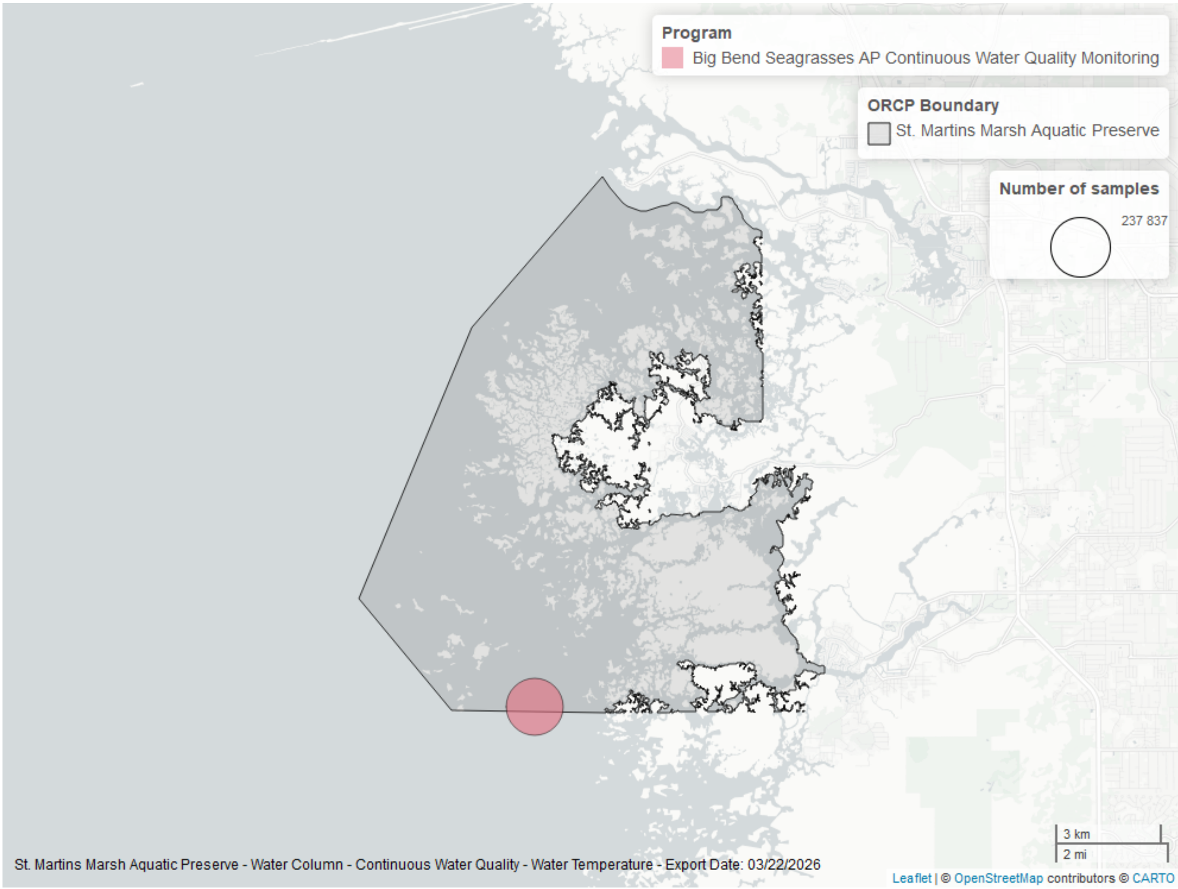

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

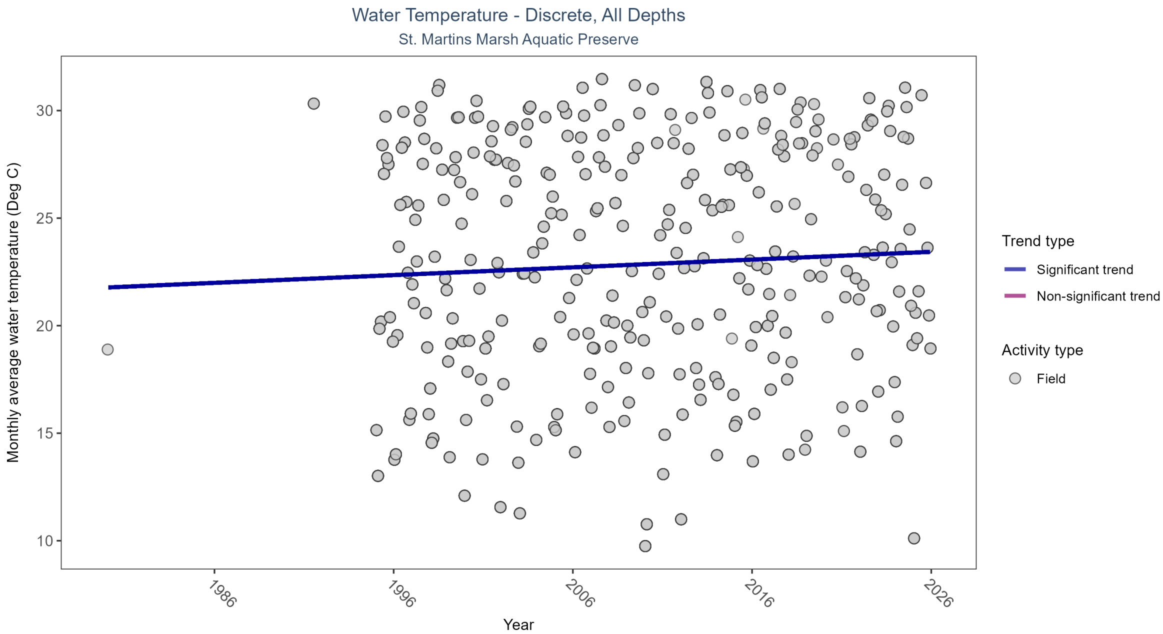

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	8501	33	1980 - 2025	22.6	0.11732	21.7742	0.03605	0.003

Monthly average water temperature increased by 0.04°C per year.

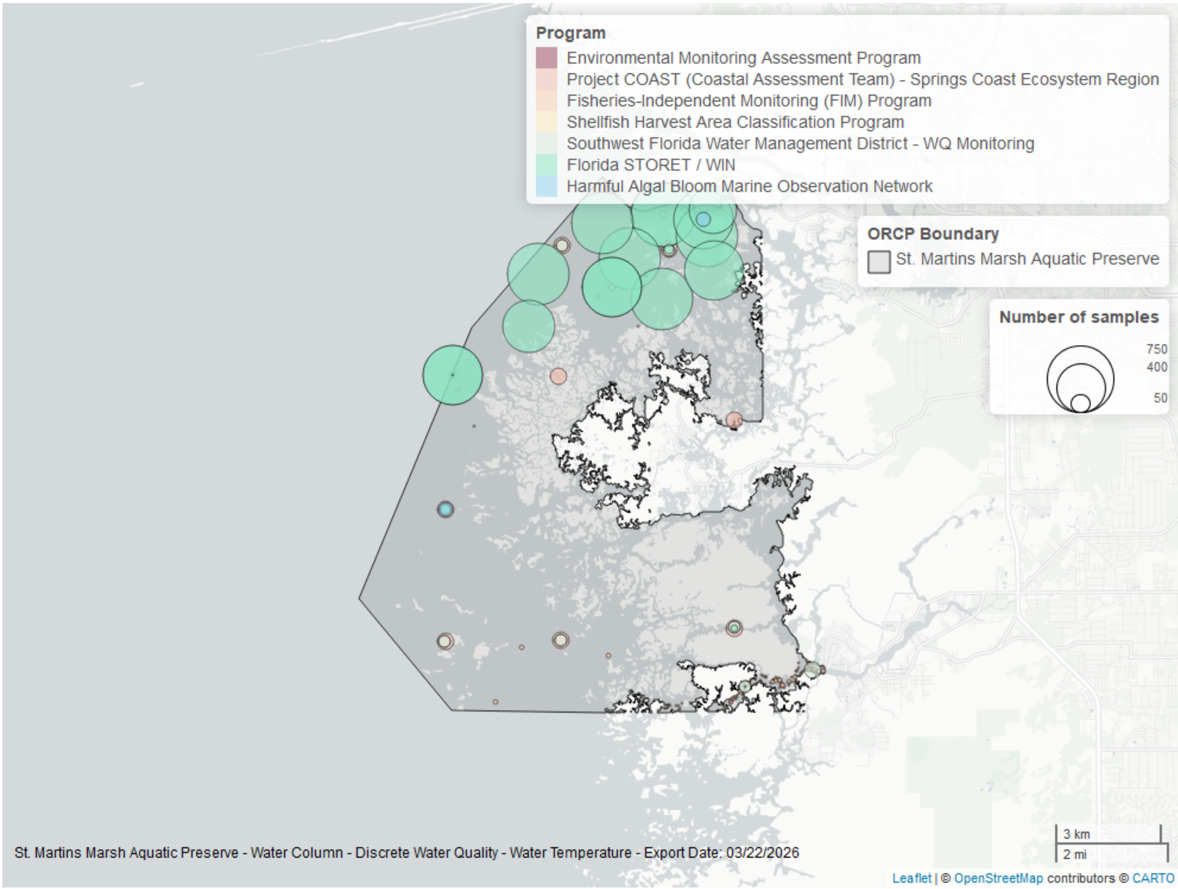

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

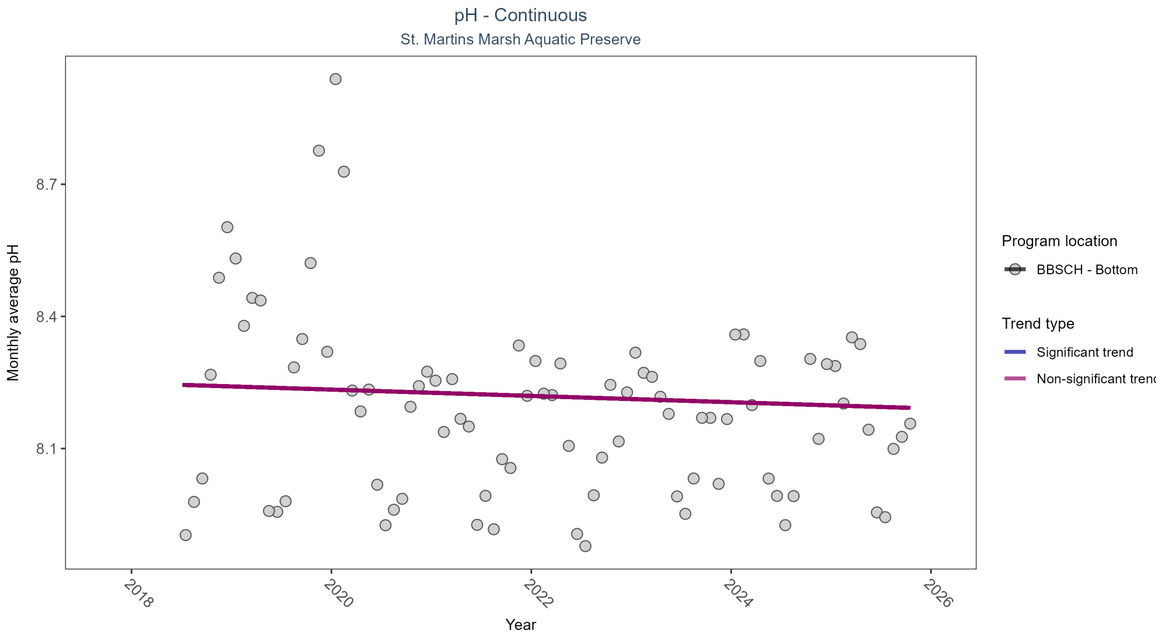

Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	No significant trend	229419	8	2018 - 2025	8.2	-0.1	8.25	-0.01	0.2865

No detectable change in monthly average pH was observed at one location.

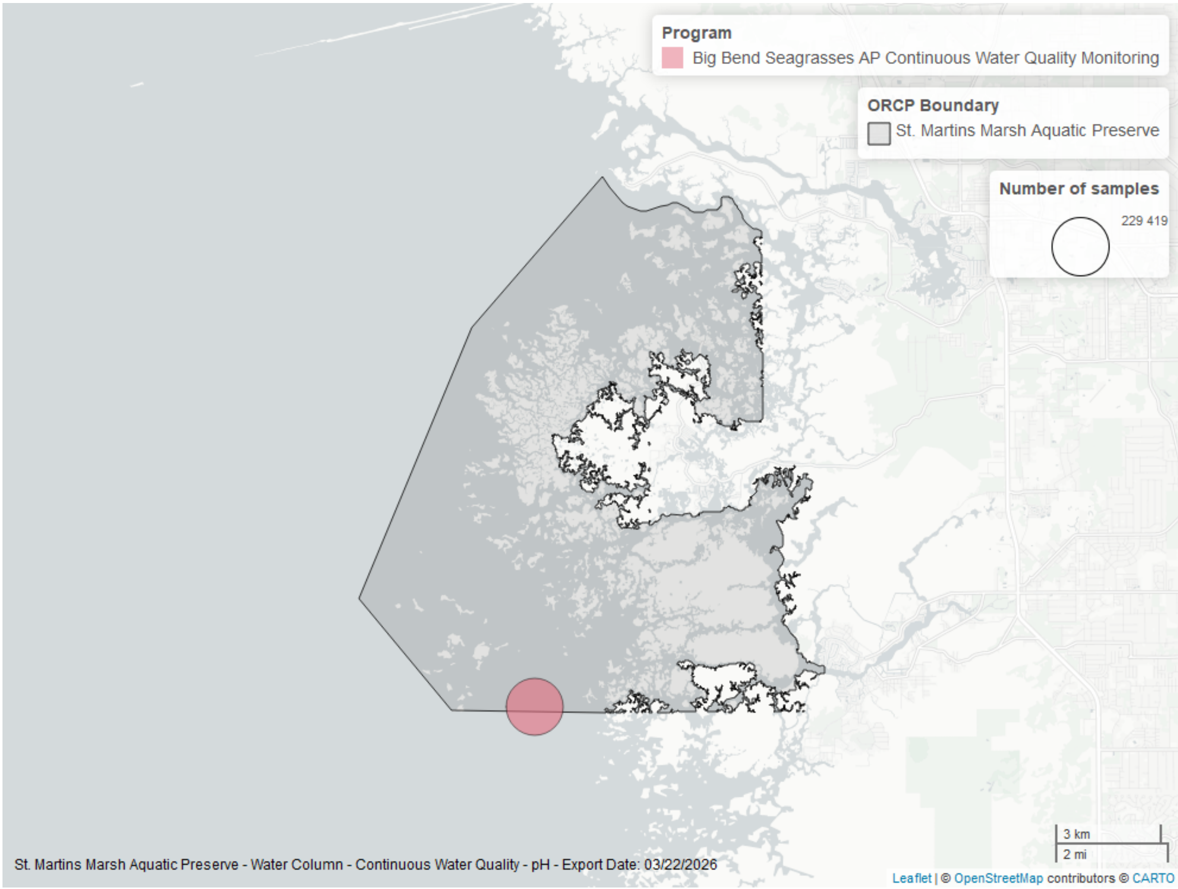

Figure 22: Map showing location of ph continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

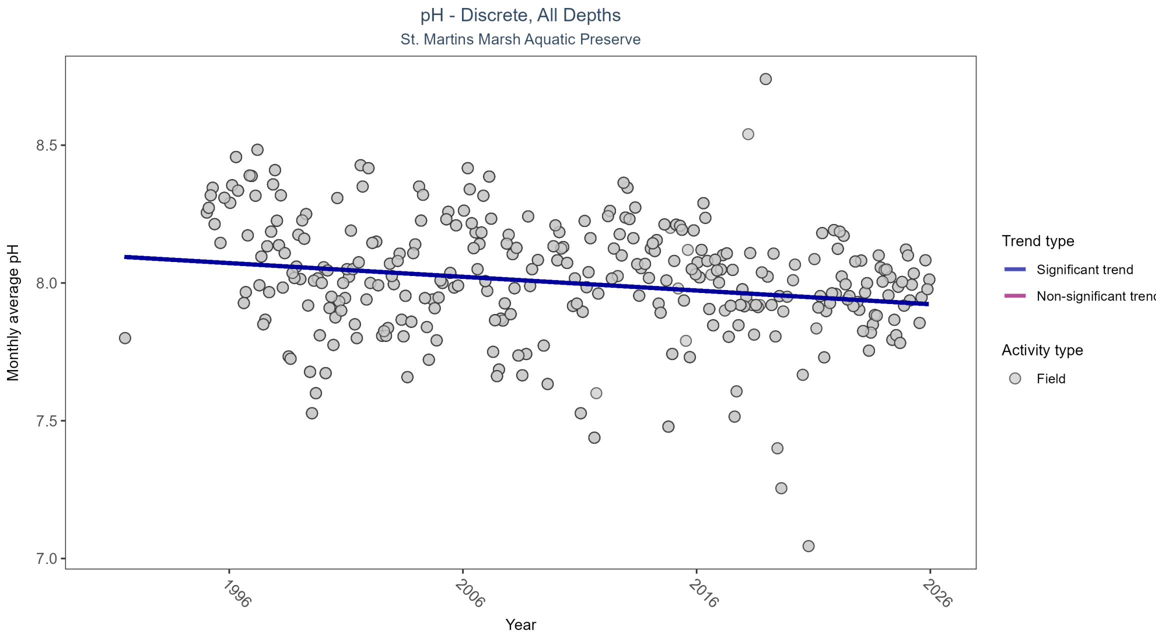

Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	4421	32	1991 - 2025	8.09	-0.16693	8.09704	-0.00498	0

Monthly average pH decreased by less than 0.01 pH units per year.

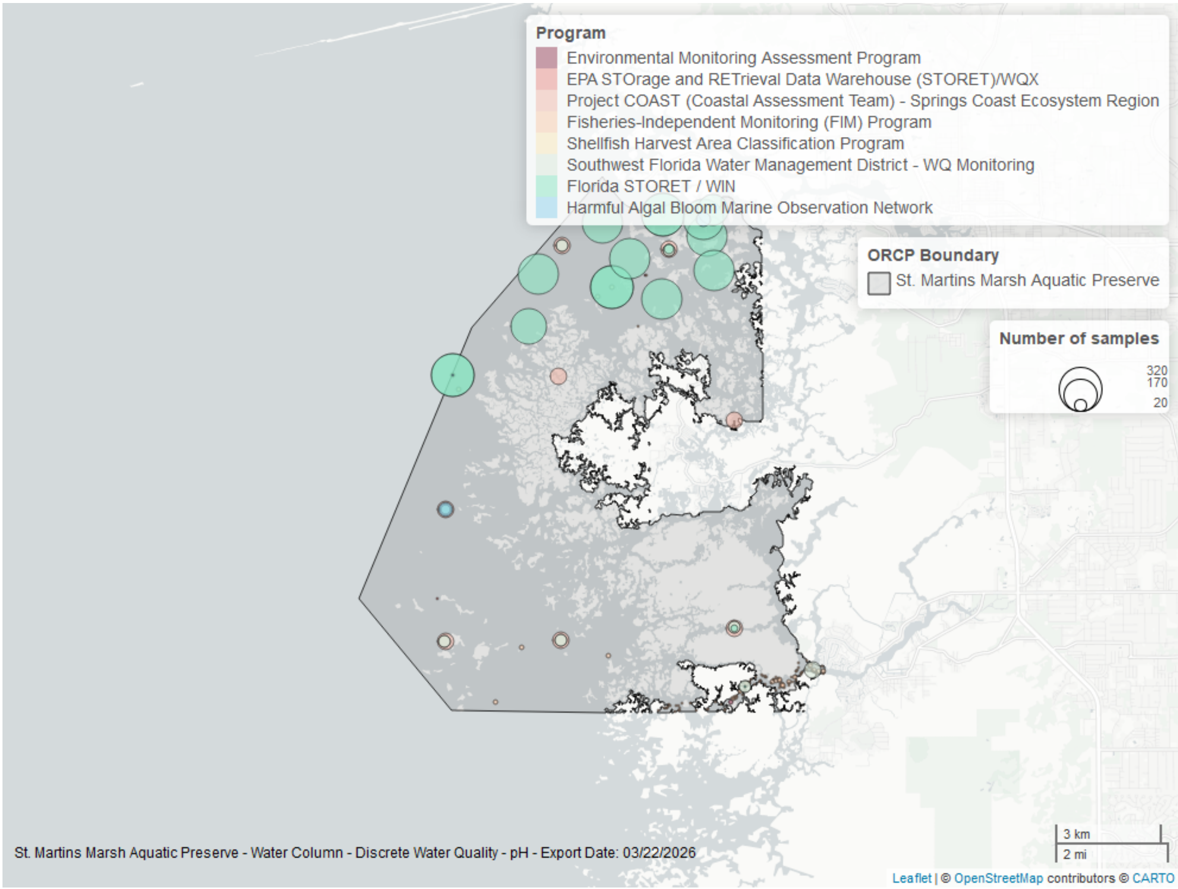

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

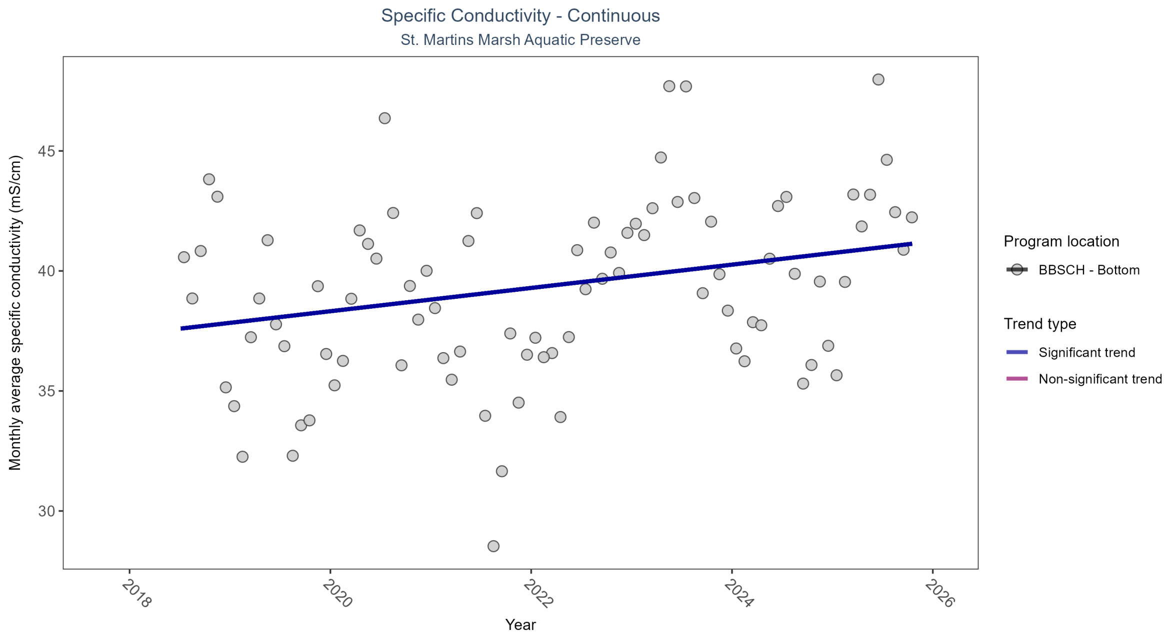

Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly increasing trend	234051	8	2018 - 2025	39.55	0.27	37.35	0.49	0.0025

At one program location, monthly average specific conductivity increased by 0.49 ms/cm per year.

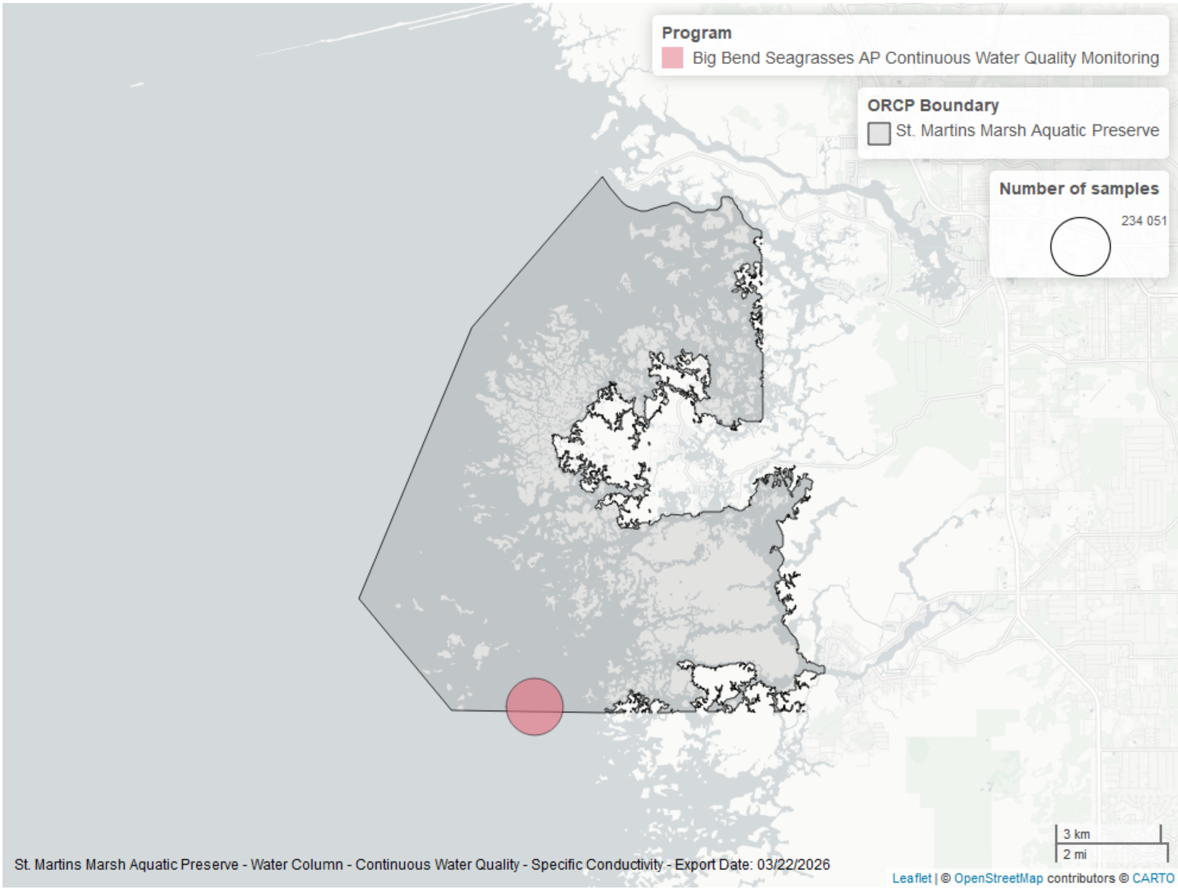

Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

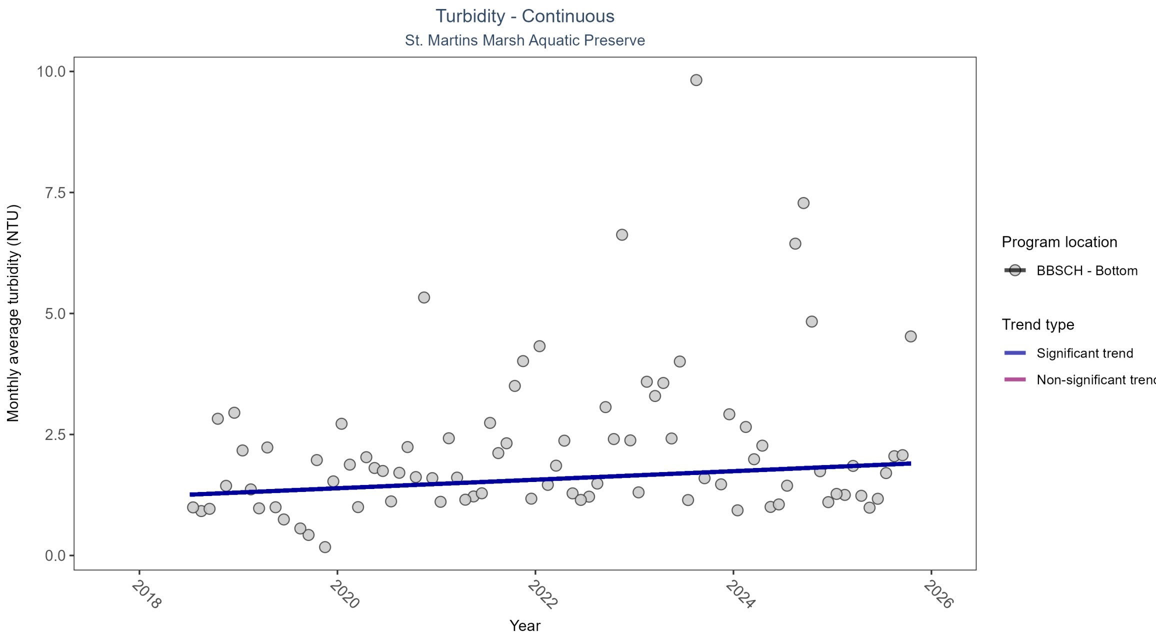

Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCH	Significantly increasing trend	204374	8	2018 - 2025	1	0.19	1.21	0.09	0.027

At one program location, monthly average turbidity increased by 0.09 NTU per year.

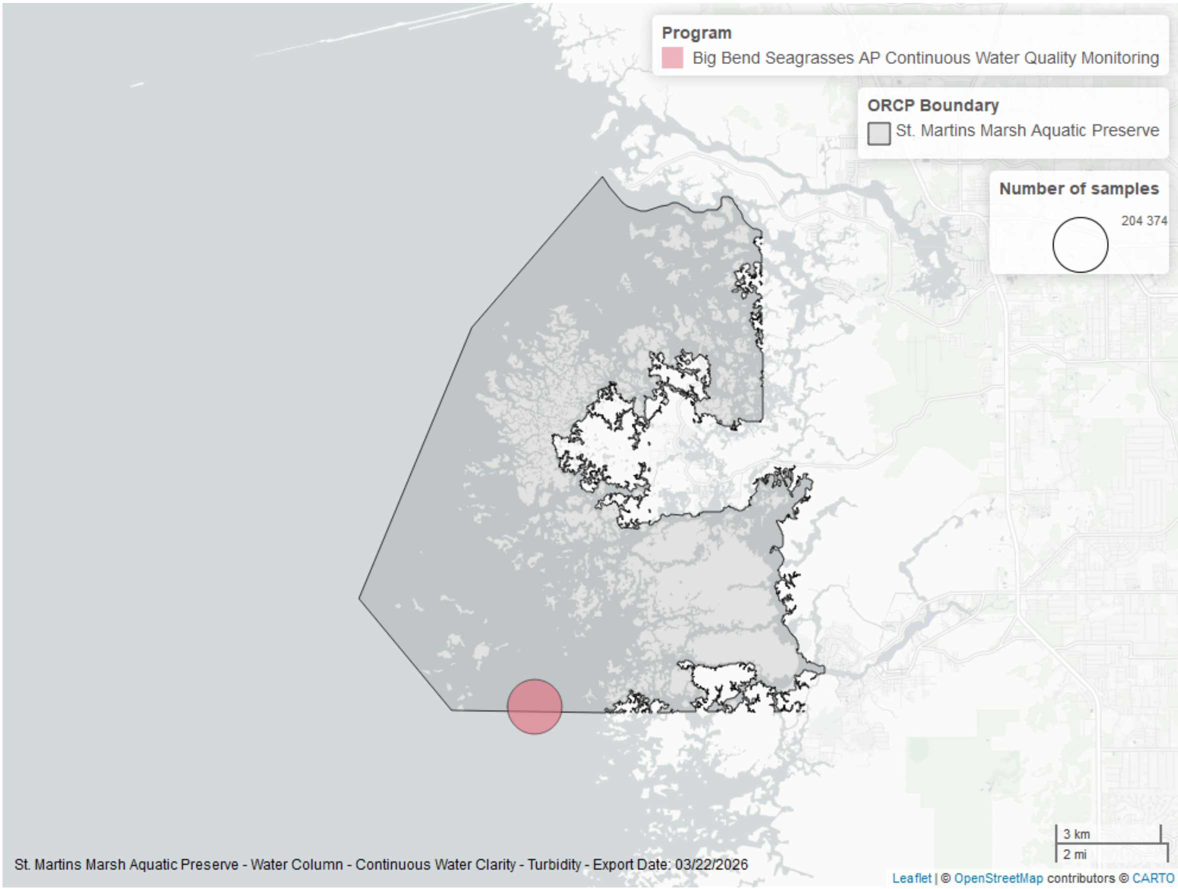

Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

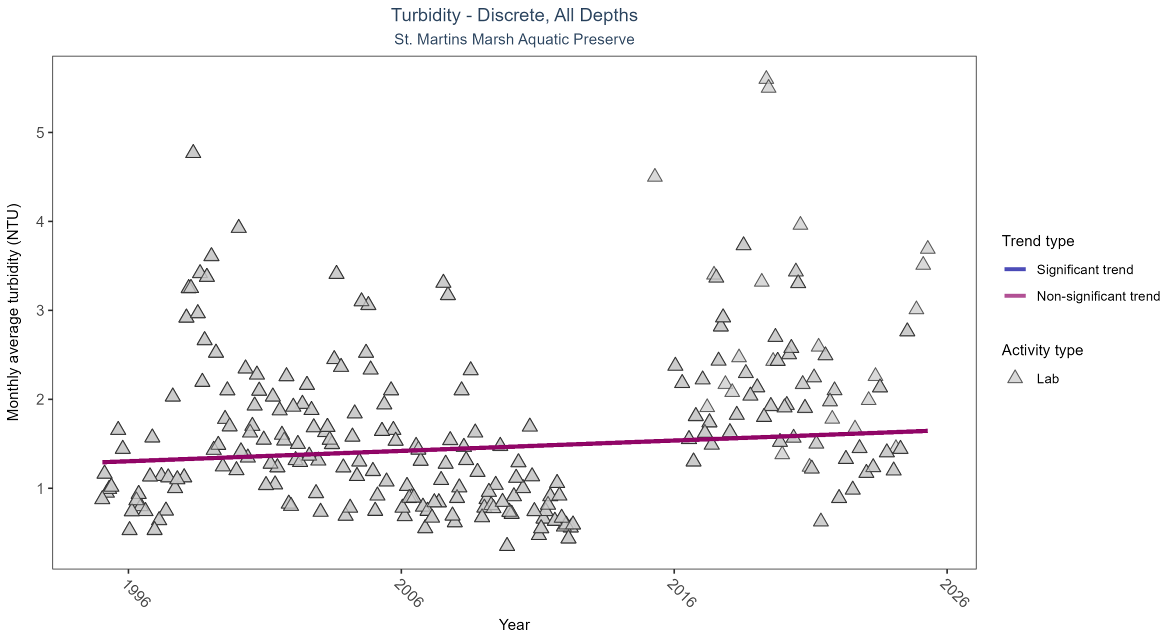

Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	3083	29	1995 - 2025	1.1	0.07458	1.29219	0.01165	0.1002

Turbidity showed no detectable trend between 1995 and 2025.

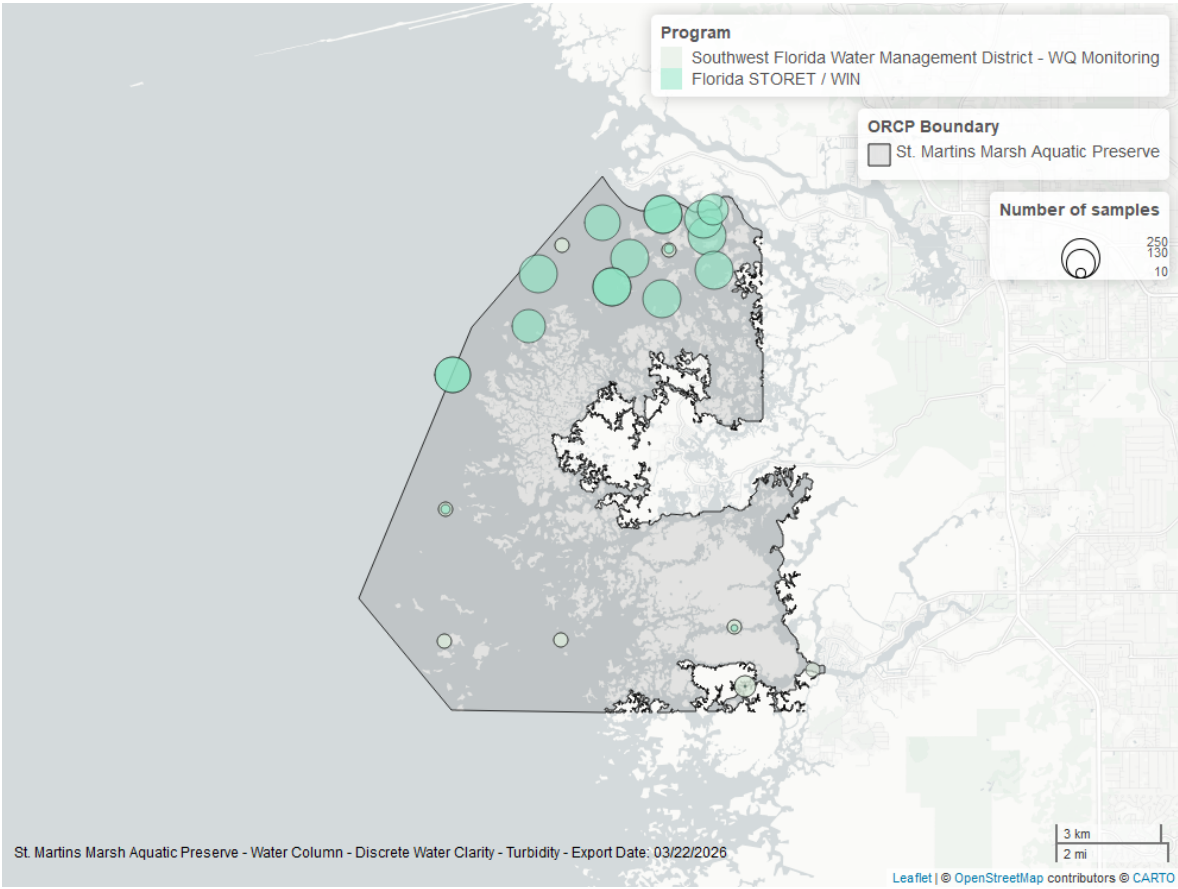

Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

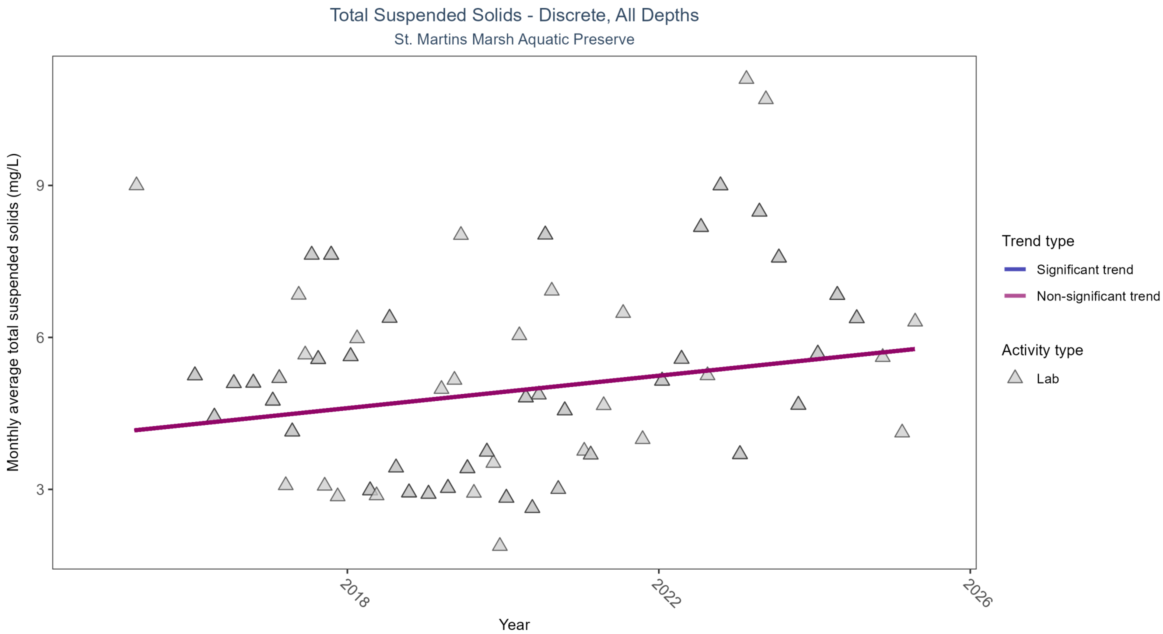

Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	256	11	2015 - 2025	4.63	0.14354	4.12333	0.16009	0.1693

Total suspended solids showed no detectable trend between 2015 and 2025.

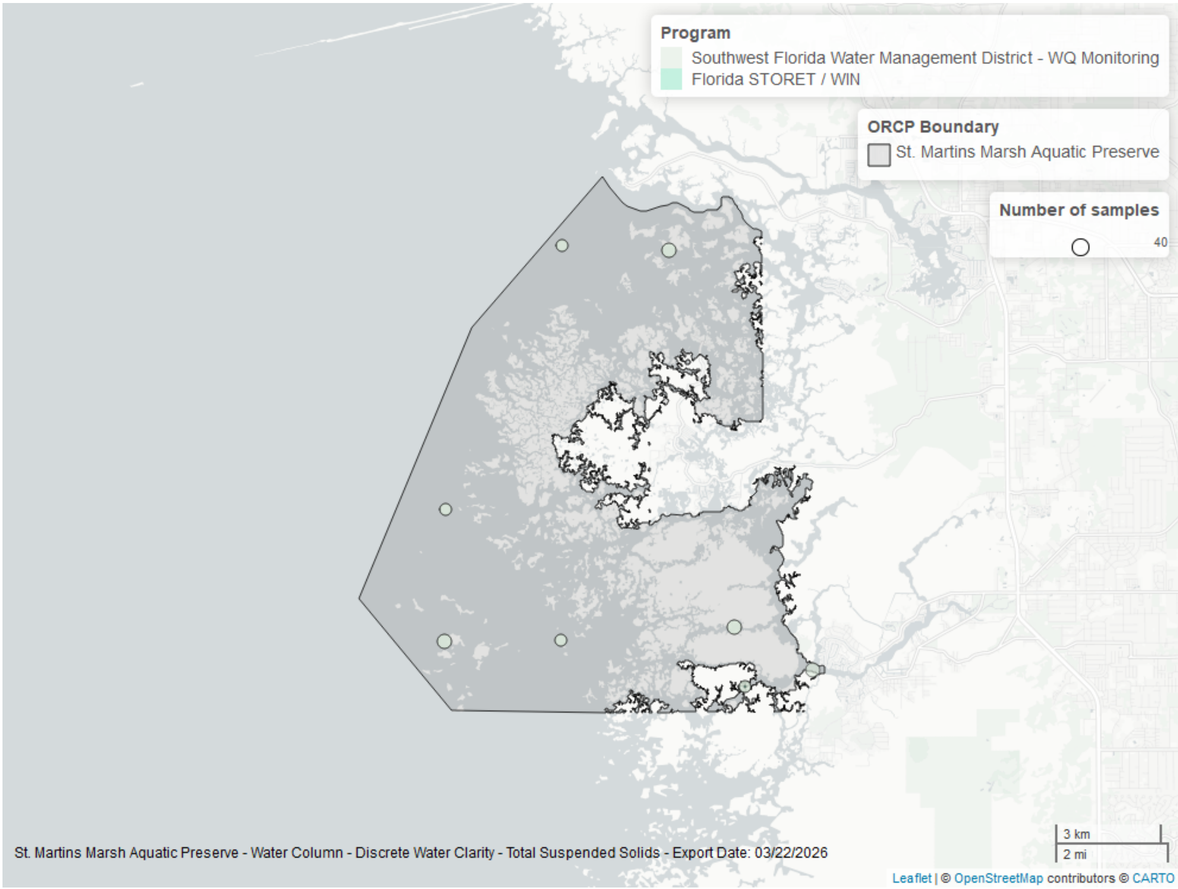

Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

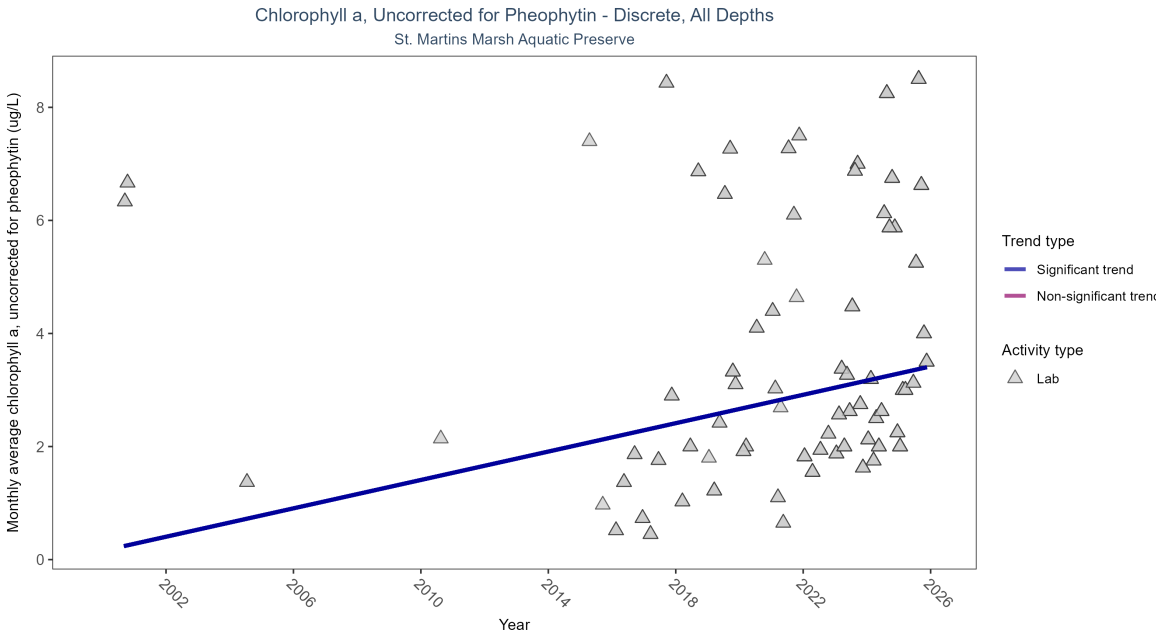

Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	384	14	2000 - 2025	2.225	0.31679	0.15318	0.12556	0.0107

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.13 µg/L per year, indicating a decrease in water clarity.

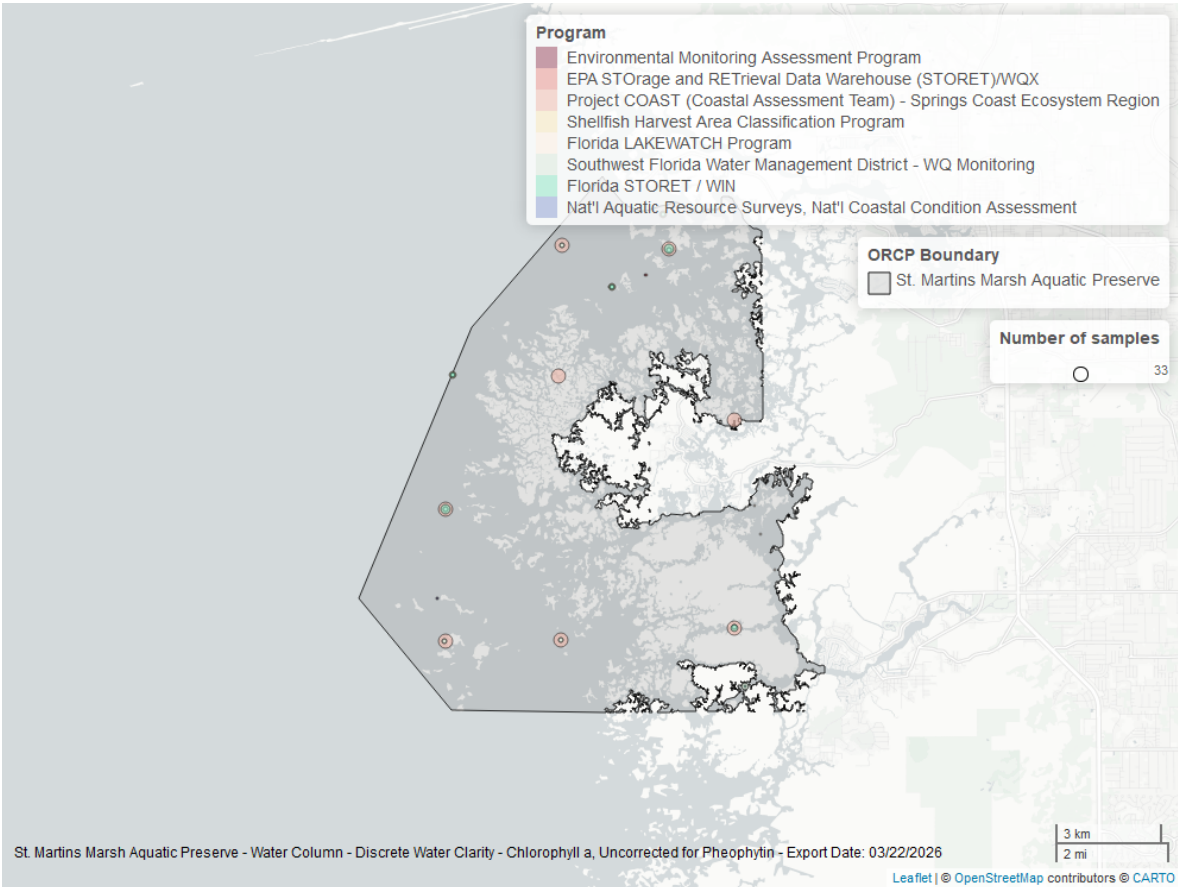

Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

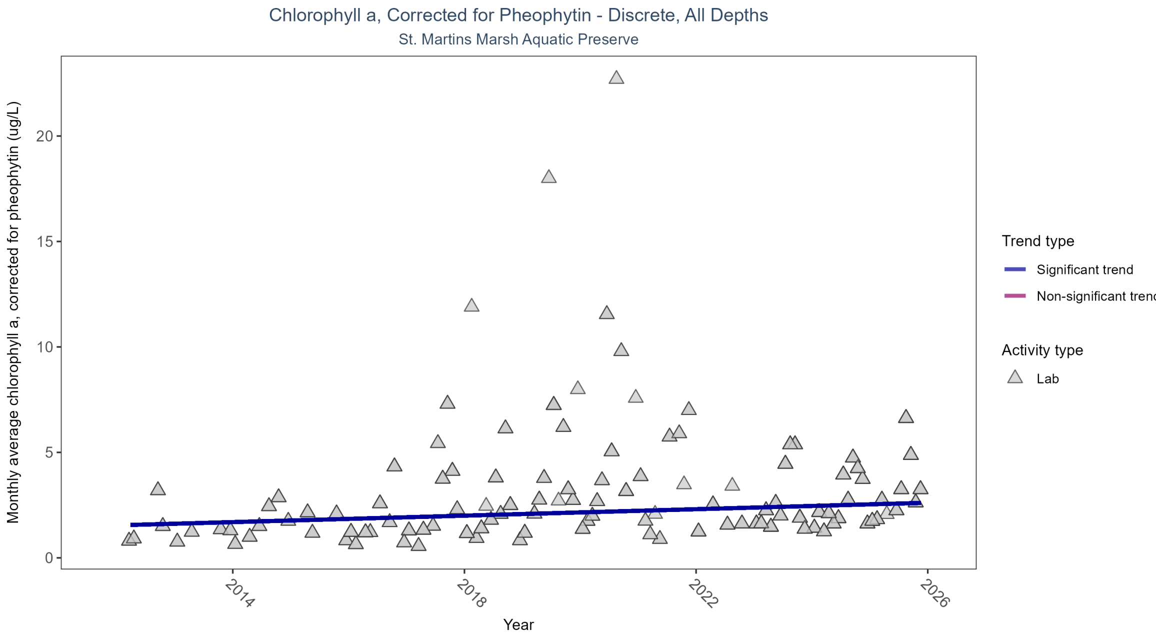

Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	635	14	2012 - 2025	1.8	0.27736	1.54162	0.07657	1e-04

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.08 µg/L per year, indicating a decrease in water clarity.

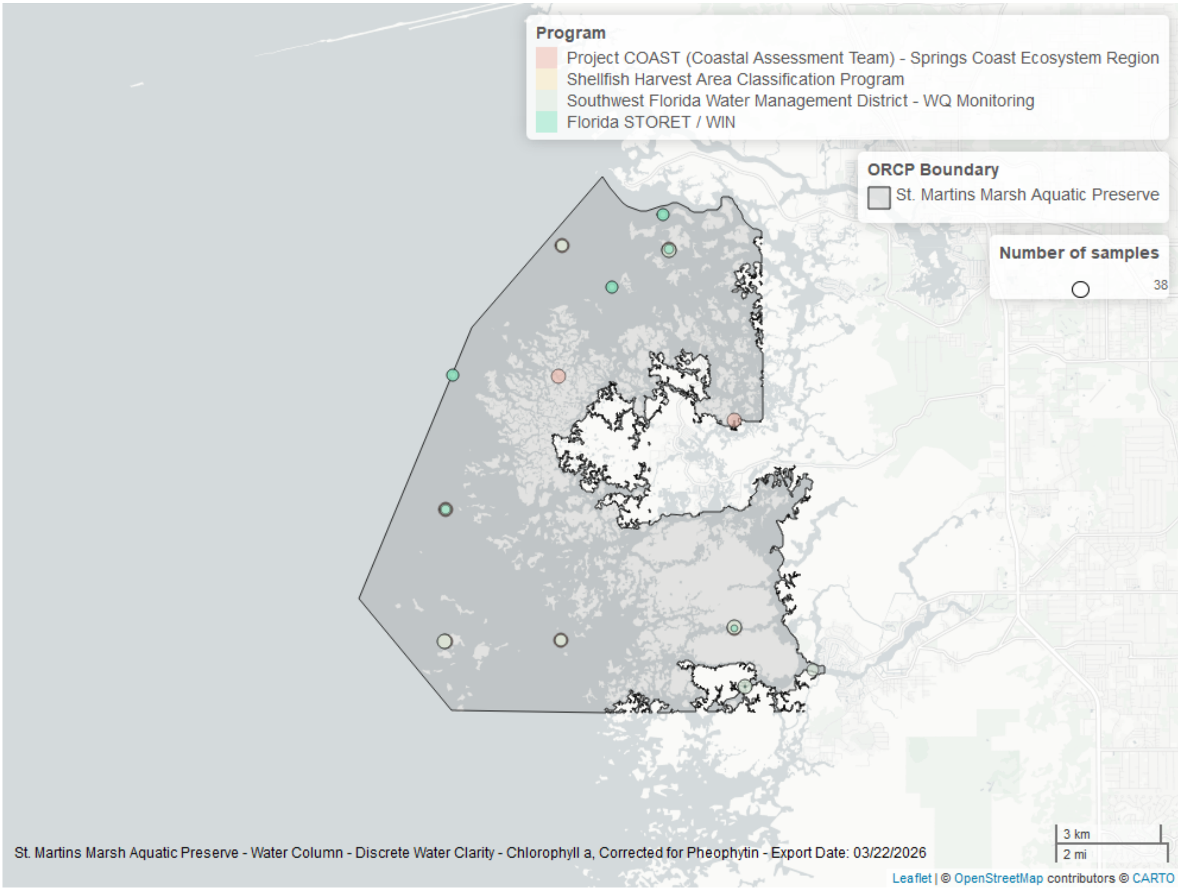

Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

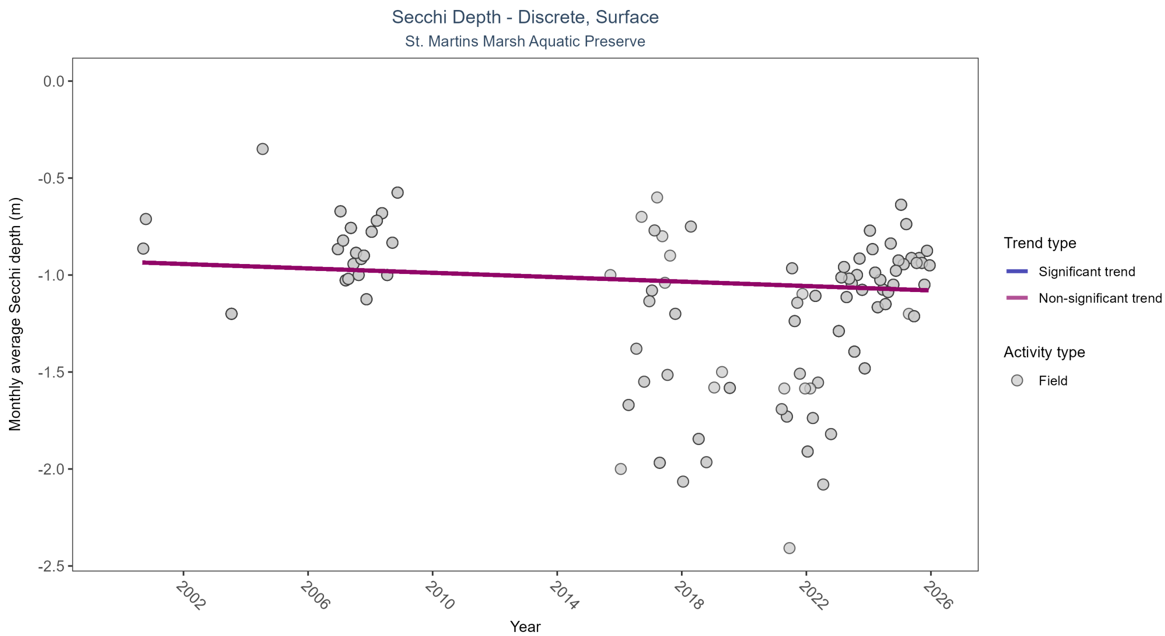

Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	582	16	2000 - 2025	-1	-0.09962	-0.93175	-0.00568	0.2809

Monthly average Secchi depth became deeper by 0.01 m per year, indicating an increase in water clarity.

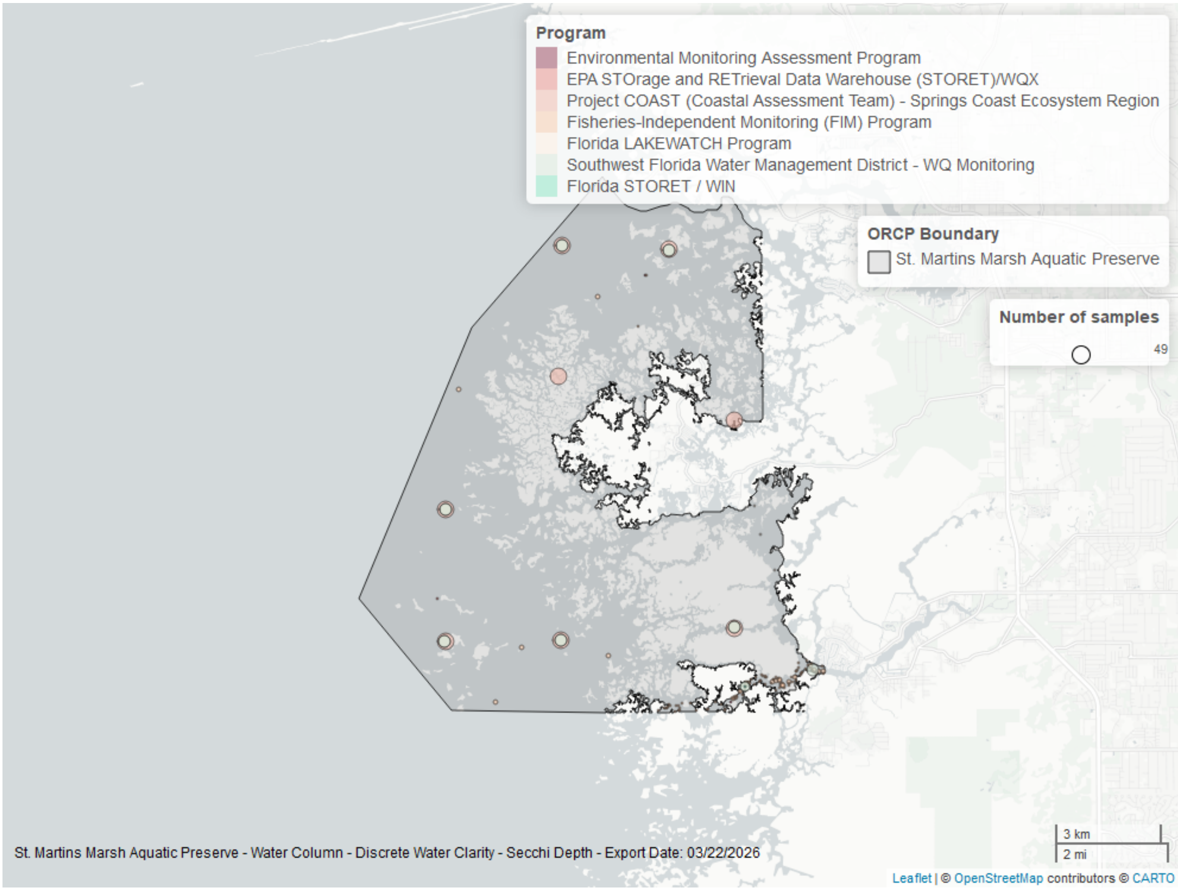

Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

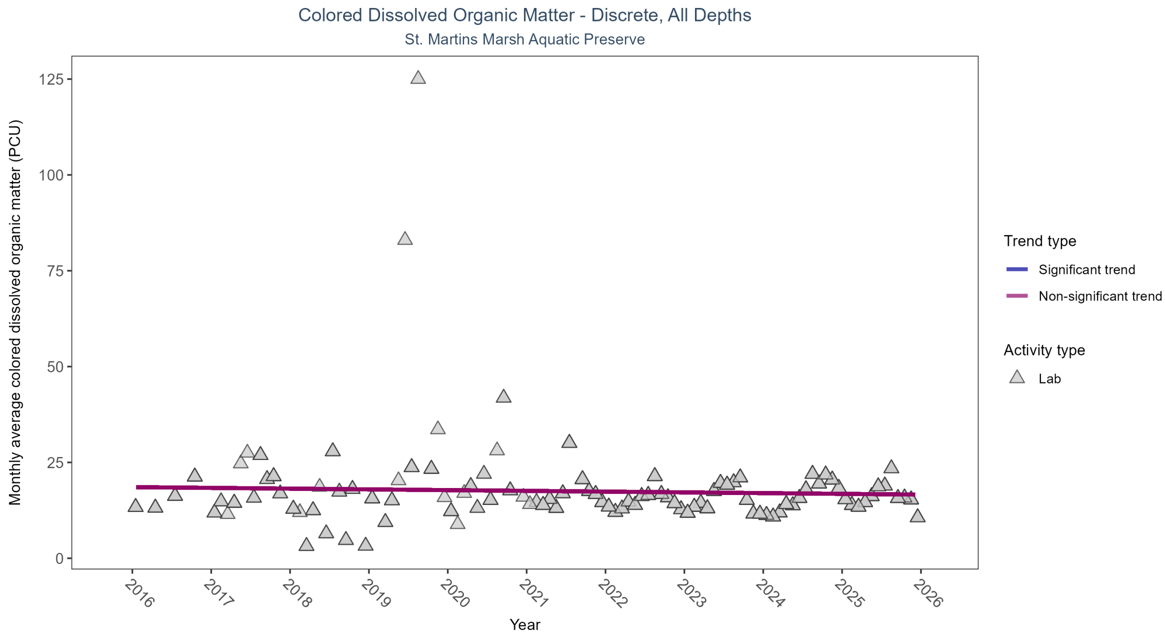

Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	731	10	2016 - 2025	13.9	-0.1162	18.54988	-0.19459	0.1619

Colored dissolved organic matter showed no detectable trend between 2016 and 2025.

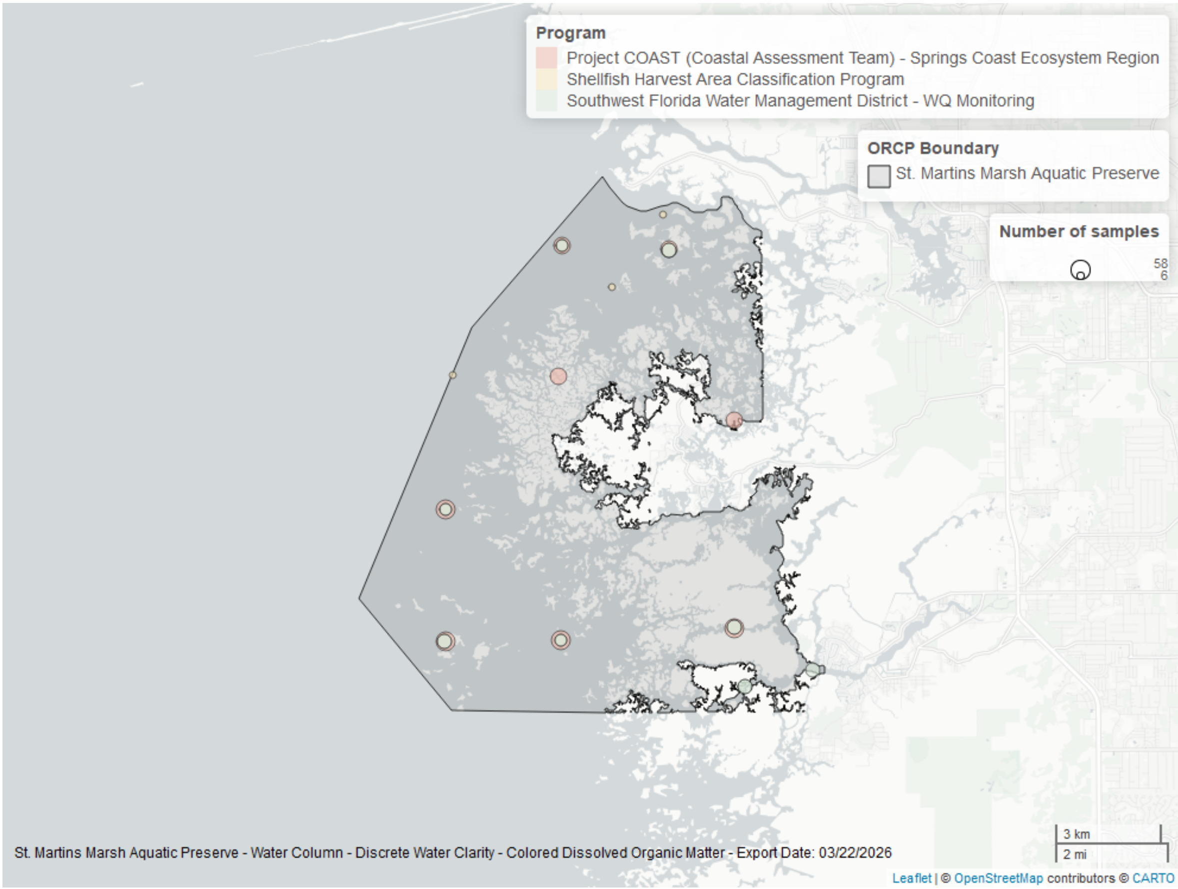

Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.